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 SURVEY

Student's Engagement Detection Based on Computer Vision: A Systematic Literature Review

IKRAM QARBAL¹, NAWAL SAEL¹, AND SARA OUAHABI¹

Laboratory of Information Technology and Modeling, Faculty of Sciences Ben M'Sick, Hassan II University of Casablanca, Casablanca 20000, Morocco

Corresponding author: Ikram Qarbal (ikram.qarbal-etu@etu.univh2c.ma)

ABSTRACT Student engagement detection is receiving increasing attention due to its crucial role in influencing academic performance and learning outcomes. As learning environments increasingly shift toward online and hybrid modes, understanding and monitoring student engagement through automated means has become an essential focus in educational technology. Computer vision techniques offer promising capabilities for detecting and interpreting student behaviors indicative of engagement. This paper presents a systematic literature review (SLR) on engagement detection in learning environments using computer vision methods. The objective of this study is to examine and categorize the types of student engagement detected, the learning contexts in which detection occurs, and the methods employed for such detection. Specifically, we analyze the types of data sources, datasets, and features used, as well as the preprocessing and feature engineering techniques applied to enhance model accuracy. Furthermore, we investigate the types of machine learning/Deep learning models adopted and how their performance is evaluated. Based on the findings from selected studies, this review aims to identify key contributions, existing challenges, and potential directions for future research in the domain of automated student engagement detection.

INDEX TERMS Computer vision, deep learning, machine learning, student engagement detection, systematic literature review.

I. INTRODUCTION

Student engagement stands as the vital determinant which shapes both educational experience quality and academic results [1]. Student engagement defines how students actively participate in educational activities while determining their academic outcomes and memory of information as well as their motivation levels. Student engagement assessment traditionally relied on manual observations along with self-reported measures which struggle to maintain objectivity and scalability [1]. Educational environments have significantly transformed with the integration of technology [2], leading institutions to seek more efficient and real-time methods to assess student engagement.

Different engagement detection methods have developed concurrently with advances in technology throughout the

years. Subjective human-based manual techniques provided informative yet unreliable results at early stages of these evaluation approaches. Semi-automatic tools such as log-file analysis [3] from learning management systems started appearing as a new option by measuring student engagement through completion rates and task interaction patterns. These assessment techniques failed to detect important behavioral and emotional dimensions of student engagement even though they provided some useful information.

The recent progress in artificial intelligence especially in computer vision has considerably boosted engagement detection capabilities through automated analysis of non-verbal expressions including facial expressions and gaze direction as well as gestures and posture [4]. The widespread shift to remote and hybrid learning during the COVID-19 pandemic further accelerated the adoption of these technologies [5], as educators faced the challenge of maintaining student engagement in virtual classrooms. Since its introduction

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computer vision has evolved into a fundamental engagement detection tool [6] providing both objective and scalable solutions which function across a range of learning spaces from digital platforms to smart classrooms.

The detection of engagement continues to be an intricate issue even with current developments. The construct of engagement incorporates multiple elements which manifest differently in behavioral and emotional and cognitive ways across distinct social contexts and age brackets and cultural traditions [7]. The detection approaches for engagement differ between learning environments since facial expressions [8] along with gaze tracking work best for virtual spaces whereas smart classrooms utilize gesture and posture methods. The differences between contexts emphasize our requirement for better knowledge about how to use and assess engagement detection methods effectively. The process of SLRs [9] helps researchers to integrate knowledge while showing trends for guiding upcoming research activities. Researchers apply organized evaluation models to existing research to produce transparent findings and promote duplicable analysis of difficult scientific domains.

An SLR method consolidates dispersed research findings which generates a comprehensive understanding about new technologies and approaches in student engagement detection. Researchers have conducted various SLRs on similar subject matter which yielded important revelations for each area. Subramanian and Mahmoud [10], arranged engagement studies into distinct areas that covered dimensions of engagement as well as influencing factors and measurement approaches. The authors created a baseline taxonomy to study engagement that included surveys and project-based learning as traditional methods. The review examined computational methods but they did not receive primary emphasis because it surveyed a wide array of topics.

In [11], Anwar et al. focused on sentiment analysis of online education through the exploration of text and facial emotion identification techniques for student emotional assessment. An evaluation of various techniques including lexicon-based methods and machine learning models demonstrated the essential importance of emotions in the learning process during the review. The main emphasis on sentiment analysis in this review narrowed down the exploration of engagement but provided minimal insights into a comprehensive understanding of the construct.

Ilić et al. in [12] examined intelligent techniques in e-learning, including knowledge tracing, educational data mining, and adaptive assessments. This review showcased the transformative potential of AI-driven approaches to personalize learning experiences. While its broad scope covered various intelligent systems, the specific focus on engagement detection techniques, particularly those leveraging computer vision, was less pronounced.

These prior reviews have provided foundational insights into engagement research and computational techniques, but there remains an opportunity to delve deeper into the

intersection of student engagement detection and computer vision. This study aims to build on these works by offering a focused and up-to-date SLR on engagement detection using computer vision, addressing methods, datasets, evaluation metrics, and challenges. By consolidating and expanding upon prior research, this review seeks to contribute to the growing body of knowledge and guide future developments in the field.

The remainder of this paper is structured as follows. Section II reviews related work to situate this study within the broader research landscape. Section III outlines the methodology, including search strategies and study selection criteria. Section IV presents the findings, addressing the research questions. Section V discusses the key trends, challenges, and opportunities identified. Finally, Section VI concludes the study with insights and directions for future work.

II. RELATED WORKS

Student engagement detection has gained significant attention in educational research, particularly with advancements in computer vision-based methods. Unlike traditional self-reports or survey-based techniques, computer vision enables non-intrusive, real-time engagement analysis by extracting behavioral and affective cues from students during learning activities. The primary existing research investigates four main visual aspects including facial expression analysis and gaze tracking together with head and body posture examination as well as their fusion techniques. As part of this related work, the following sections examine these core techniques alongside temporal modeling approaches, real-time deployment strategies, and data-centric solutions, outlining major advancements and current research gaps.

A. COMPUTER VISION-BASED ENGAGEMENT DETECTION TECHNIQUES

The field of computer vision-based engagement detection has been primarily driven by visual modalities such as facial expressions, eye gaze, head pose, and body posture. Among these, facial expression recognition (FER) has received the most attention, serving as a proxy for inferring emotional and cognitive engagement. Deep learning approaches, particularly convolutional neural networks (CNNs), have shown strong performance in classifying engagement levels based on facial cues, with models achieving accuracies above 90% in several studies (e.g., Gupta et al. [13], Bhardwaj et al. [14], Lima et al. [15]). These systems typically rely on datasets like FER-2013, CK+, and DAiSEE, using expressions such as happiness, surprise, or neutrality to infer learner attentiveness.

In parallel, eye gaze tracking has proven effective in identifying visual attention and detecting signs of distraction. Systems using gaze direction, fixation duration, and blinking patterns have reported high validation accuracies, for example, Ahmed et al. [16] achieved 99% accuracy using a MobileNet model trained on eye images. Webcam-based

solutions, such as those proposed by Uçar and Özdemir [17] and Wang et al. [18], offer cost-effective and scalable alternatives to hardware-dependent eye-tracking tools, with recognition rates exceeding 90% in real-time setups.

Head pose and posture estimation also contribute valuable information, especially for behavioral engagement assessment. Head movements like tilting, nodding, or turning away from the screen are often indicative of disengagement or fatigue. Studies incorporating these features have used algorithms such as Haar Cascades, HOG, and SVM classifiers to achieve solid results, for instance, Alrawahneh and Safei [19] reported an 80% agreement between their head-pose-based model and expert assessments. Postural analysis, on the other hand, is commonly used in 1:1 learning or smart classroom contexts, where body orientation and upper-body dynamics can signal shifts in engagement (Aslan et al. [20]).

Recognizing the limitations of single-modality systems, many researchers have moved toward multimodal frameworks that integrate multiple visual cues. These systems often combine facial expressions, gaze tracking, head pose, and posture analysis to construct a richer understanding of engagement. For example, Gupta et al. [21] developed a multimodal engagement index combining emotion recognition, eye blinking, and head movement, achieving 92.58% accuracy. Similarly, Qi et al. [22] proposed a cascaded model that integrates gaze, facial expressions, and action recognition, outperforming baseline models like LRCN and C3D. In another study, Kaur et al. [23] employed a deep multiple-instance learning framework to process facial and bodily cues across time, enabling engagement localization with improved generalization to real-world conditions.

These computer vision-based systems demonstrate promising results, with several studies reporting classification accuracies in the range of 85–99%. Their real-time capabilities, coupled with advances in lightweight model architectures, have made them increasingly suitable for deployment in online learning platforms and intelligent tutoring systems. However, challenges remain in handling variable lighting conditions, occlusions, and user diversity, factors that have motivated further exploration of temporal modeling, data augmentation, and transfer learning strategies.

B. TEMPORAL AND SEQUENTIAL MODELING

Since engagement is inherently dynamic, temporal modeling has become increasingly important for capturing changes in attention and affective states over time. Rather than relying on isolated frames, many recent approaches apply sequential deep learning architectures such as Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), or Transformer-based models to analyze video sequences of learners.

For instance, Buono et al. [8] used LSTM networks to model facial action units, gaze behavior, and head movements over time, finding a strong correlation between predicted engagement levels and self-reported outcomes. Similarly,

Chang et al. [24] applied an attention pooling mechanism over temporal sequences of posture and facial cues, achieving a significant reduction in mean squared error (MSE) compared to baseline models. Ai et al. [25] introduced a video transformer architecture that outperformed previous models on EmotiW-EP and DAiSEE datasets, achieving MSE scores of 0.0495 and 0.0377 respectively.

Sequential models not only offer improved performance but also enable more responsive and adaptive learning systems, as they can monitor ongoing engagement trends rather than static snapshots. This is especially useful for applications in MOOCs, asynchronous video lectures, and real-time tutoring, where learners' attention may fluctuate significantly throughout a session.

C. REAL-TIME AND EDGE-OPTIMIZED SYSTEMS

As the deployment of engagement detection systems expands beyond research environments, the need for real-time performance and hardware efficiency has become increasingly prominent. Many recent studies have shifted focus toward developing lightweight, scalable models that can run on edge devices such as laptops, tablets, or smartphones without the need for cloud infrastructure.

Boulanger et al. [26] introduced an interpretable, edge-compatible model using SqueezeNet, which significantly reduced parameter count while maintaining over 93% accuracy. Harb et al. [27] achieved 86.29% accuracy on a specialized dataset using a deep emotion recognition model optimized for edge deployment. Likewise, Wang et al. [18] demonstrated that cost-efficient systems based on standard webcams can outperform commercial eye-trackers in engagement detection when supported by robust deep learning pipelines.

These models often strike a balance between latency, energy consumption, and inference quality, making them ideal for large-scale deployment in educational settings where infrastructure constraints are common. However, their reduced complexity may come at the cost of interpretability or adaptability across diverse learner populations.

D. DATASETS, LABELING STRATEGIES, AND TRANSFER LEARNING

Progress in engagement detection has been closely tied to the availability and quality of datasets. Public datasets such as DAiSEE, EmotiW, and FER-2013 have served as benchmarks for model evaluation, but many researchers have developed custom datasets to address limitations in sample diversity, label consistency, or real-world variability. For instance, Kaur et al. [23] introduced the Engagement in the Wild dataset with 264 videos from 91 participants in uncontrolled environments, allowing for more realistic model training.

Given the challenges of collecting large, labeled datasets, transfer learning has become a popular strategy. Nezami et al. [28] applied transfer learning by pretraining on FER-2013 and fine-tuning on a smaller engagement dataset,

improving accuracy to 72.38% and AUC to 73.74%. Similarly, Watanabe et al. [29] utilized role-acting data collection combined with pre-trained models to avoid the limitations of self-reporting.

To further address labeling inconsistencies, some works have proposed weakly supervised or semi-supervised learning techniques (e.g., Kaur et al. [23], allowing systems to learn engagement intensity across time without frame-level annotations. Others such as Abedi et al. [30], Psaltis et al. [31] explore ordinal classification or emotion clustering to better capture the nuanced nature of engagement states.

III. RESEARCH METHODOLOGY

A literature review is a fundamental component of academic research, providing a foundation for understanding and synthesizing existing knowledge. In this review, we employ the SLR's technique to study and analyze student engagement detection methods using computer vision. This approach enables us to explore engagement classification techniques, data sources, features, preprocessing steps, and algorithms in a structured and comprehensive manner. An SLR systematically identifies, selects, and critically evaluates research to address a set of predefined questions. We organized, conducted, and reported this review following established SLR guidelines outlined by Kitchenham et al. [32] to ensure rigor and transparency.

A. RESEARCH QUESTIONS

The main research questions (RQs) that need to be answered in order to conduct the SLR for the suggested study are as follows:

RQ1: What are the types of student engagement, and corresponding detection Methods used in different learning environments?

RQ2: What datasets and features are used for detecting student engagement?

RQ3: What preprocessing and feature engineering techniques are applied to improve the quality and accuracy of student engagement detection models?

RQ4: What types of computer vision models are used to detect student engagement, and how is their performance evaluated?

B. SEARCH STRATEGY

To ensure a comprehensive and systematic review of the literature, a search strategy was devised to identify relevant studies focusing on student engagement detection using computer vision. The search was conducted across four prominent databases: IEEE Xplore, SpringerLink, Google Scholar and Scopus, as these platforms host a wide range of peer-reviewed publications in computer science, and educational technology.

The search query was carefully constructed using a combination of keywords and Boolean operators to capture studies addressing various aspects of engagement detection. The query was adjusted for each database to account for differ-

ences in search syntax while maintaining its core structure. The final query was designed to include the following terms:

- **Engagement Terminology:** “student engagement”, “learner engagement”, “academic engagement”, “classroom engagement”, “educational engagement”
- **Technology Focus:** “computer vision”, “image processing”, “visual analysis”, “video analysis”, “deep learning”, “machine learning”
- **Detection Focus:** “detection”, “monitoring”, “assessment”, “recognition”, “classification”, “prediction”, “tracking”
- **Engagement Cues:** “emotion recognition”, “gaze tracking”, “facial expression”, “head pose”, “body posture”, “eye tracking”, “affective computing”

Boolean operators were used to combine these blocks, and each query was adjusted to match the syntax of the target database.

The final queries used were as follows:

IEEE Xplore and SpringerLink:

(“student engagement” OR “learner engagement” OR “academic engagement” OR “classroom engagement” OR “educational engagement”)

AND (“computer vision” OR “image processing” OR “visual analysis” OR “video analysis” OR “deep learning” OR “machine learning”)

AND (“detection” OR “monitoring” OR “assessment” OR “recognition” OR “classification” OR “prediction” OR “tracking”)

AND (“emotion recognition” OR “gaze tracking” OR “facial expression” OR “head pose” OR “body posture” OR “eye tracking” OR “affective computing”)

Scopus:

TITLE-ABS-KEY((“student engagement” OR “learner engagement” OR “academic engagement” OR “classroom engagement” OR “educational engagement”)

AND (“computer vision” OR “image analysis” OR “video” OR “deep learning” OR “machine learning” OR “AI”)

AND (“detection” OR “monitoring” OR “assessment” OR “recognition” OR “classification” OR “prediction” OR “tracking”)

AND (“emotion” OR “facial” OR “gaze” OR “posture” OR “affective”))

Google Scholar:

Due to limited Boolean support, we used the advanced search form:

All words: student engagement detection, computer vision

Exact phrase: “student engagement detection”

At least one of the words: “learner engagement” “academic engagement” “classroom engagement” “deep learning” “facial expression” “gaze tracking” “emotion recognition” “head pose” “body posture” “affective computing” “visual analysis” “video analysis”

This combination of structured Boolean queries and advanced form-based filtering ensured broad coverage across disciplines while keeping results aligned with the review's

focus. Following retrieval, a systematic screening process was applied to ensure that only the most relevant and high-quality studies were included. The criteria and procedures used for study selection are detailed in the next section.

C. INCLUSION AND EXCLUSION CRITERIA

To ensure that the studies included in this review were relevant and of high quality, specific inclusion and exclusion criteria were established:

Inclusion Criteria:

- Peer-reviewed articles published in academic journals or conference proceedings.
- Articles published from 2014 to 2025
- The works published in IEEE Xplore, SpringerLink, Google Scholar and Scopus.
- Studies focusing on student engagement detection using computer vision techniques.
- Research that utilized datasets or features related to visual cues, such as facial expressions, gestures, gaze, posture, or head pose.
- Studies that reported evaluation metrics (e.g., accuracy, precision, recall, F1-score) for engagement detection models.
- Articles published in English to ensure accessibility and consistency.

Exclusion Criteria:

- Studies that focused on non-vision-based engagement detection methods (e.g., log-file analysis or text-based engagement tracing).
- Articles without accessible full texts.
- Studies that lacked sufficient details on methods, datasets, or evaluation metrics.
- Duplicate studies retrieved across multiple databases.

These criteria helped us focus the review on high-quality research directly relevant to the topic.

Next, we conducted a rigorous selection process to ensure that only relevant and high-quality studies were included in this review. We began by performing an initial search across four major databases: IEEE Xplore, SpringerLink, google Scholar and Scopus, using the query described earlier. This search yielded a total of 1387 records. Following this, we removed duplicate entries and proceeded to screen the remaining papers by carefully reviewing their titles and abstracts. At this stage, we excluded studies that were not directly related to student engagement detection or those that did not employ computer vision techniques. The next step involved a full-text review of the potentially eligible papers to ensure they aligned with our research objectives. Throughout this process, we applied a quality assessment framework to evaluate each study based on its relevance, methodological rigor, the reporting of evaluation metrics, and the novelty of its contributions. Only papers that met a minimum quality standard were retained for the final dataset. The details of the selection process are summarized in Fig. 1 using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) [33].

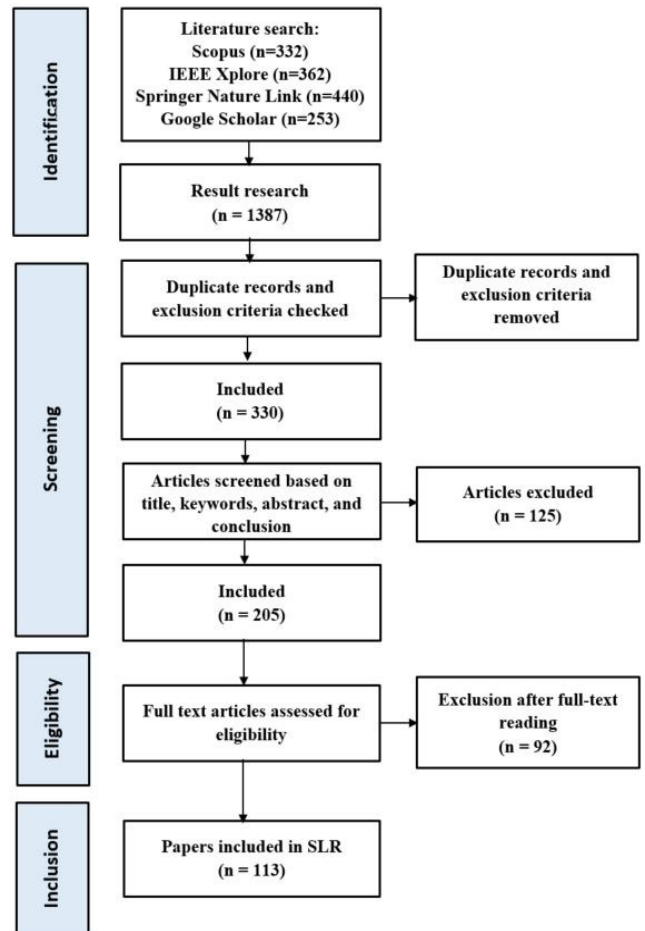


FIGURE 1. PRISMA flow diagram summarizing the article selection process for the systematic literature review.

D. DATA ANALYSIS

We identified 113 studies (Table 1) in the field of student engagement detection based on computer vision, published during the period 2014–2025. This date range was chosen based on two considerations: (1) the adoption of deep learning-based computer vision techniques in educational contexts began around 2014, and (2) the review aims to cover approximately the last decade of research, a common practice in systematic literature reviews to ensure relevance and manageability.

While computer vision has older foundations, it wasn't until the deep learning breakthrough in 2012 (e.g., AlexNet) that substantial progress was made in image-based affective computing and visual behavior analysis. These advances began to be applied to education and engagement monitoring starting around 2014, with early studies exploring facial expression recognition, head pose estimation, and gaze tracking in learning environments. Selecting 2014 as the starting point thus ensures that the review captures the modern wave of research driven by deep neural networks, which now represent the state-of-the-art in the field. The endpoint of 2025 allows inclusion of the most recent and in-progress

literature available at the time of writing. Of the 113 studies, 54 (48%) were published in conference proceedings, and 59 (52%) in journals. Table 1 presents the publication venues of the chosen studies, and Fig. 2 illustrates the annual distribution of publications over the review period.

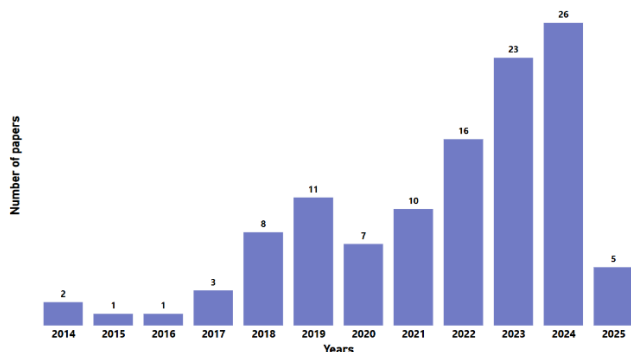


FIGURE 2. Annual distribution of selected studies on student engagement detection using computer vision (2014–2025).

To examine the 113 selected studies, we followed a structured data extraction process using Excel. For each study, we manually reviewed the full text and recorded key information into predefined columns including publication year, keywords, type of engagement addressed (emotional, behavioral, cognitive), input modalities (e.g., facial expressions, gaze, posture), machine learning techniques used, dataset characteristics, learning context (e.g., online, in-person, Virtual Reality and Augmented Reality VR/AR), and reported evaluation metrics. This structured tabulation allowed us to organize the studies thematically and quantitatively, identify common patterns, and support our analysis of trends across the literature. The themes presented in the Results section were derived directly from this tabulated data.

IV. ENGAGEMENT DETECTION: A COMPREHENSIVE OVERVIEW

In this section, we present all the answers to the research questions and the results concluded from the state of the art.

A. STUDENT ENGAGEMENT TYPES AND DETECTION METHODS IN DIFFERENT LEARNING ENVIRONMENTS (RQ1)

Student engagement is widely recognized as a pivotal factor in shaping learning outcomes, with engaged students exhibiting greater persistence, deeper comprehension, and higher performance compared to their disengaged peers.

However, engagement is not a singular construct it encompasses diverse, intertwined dimensions.

Educational researchers generally conceptualize it along three primary axes: emotional, behavioral, and cognitive engagement.

- Emotional engagement reflects the affective experiences that color the learning process, such as enjoyment, curiosity, or frustration.

- Behavioral engagement captures observable participation and task involvement, evident in attentiveness, note-taking, or classroom interaction.
- Cognitive engagement, in turn, refers to the mental effort and strategic thinking that underpin deep learning, manifesting in sustained attention, problem-solving, and reflective inquiry.

While these types are analytically distinct, in practice, they frequently overlap a student's frustration may disrupt their cognitive focus, or their physical restlessness might signal emotional disengagement.

This inherent interconnectedness has driven recent research toward multidimensional approaches, which seek to capture the complex interplay between affect, behavior, and cognition during learning.

Fig. 3 displays the research distribution between these engagement types and shows emotional engagement as the most studied dimension with 45.13% (e.g. [13], [86], [88], [94], [121]), while multidimensional approaches follow with 38.05% then cognitive engagement at 14.16% and behavioral engagement stands as the least studied type with 2.65%.

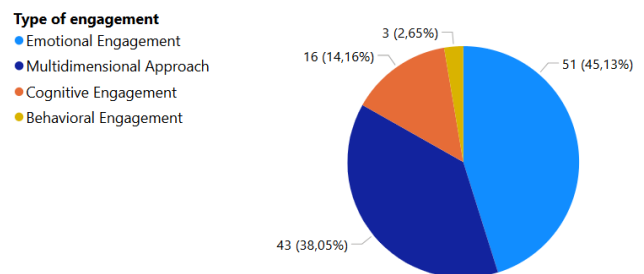


FIGURE 3. Distribution of engagement types identified in the reviewed studies. Emotional and multidimensional engagement were the most frequently reported, followed by cognitive and behavioral engagement.

While the previous section outlined the distribution of engagement types across the reviewed studies, it is equally important to understand how these types are operationalized and detected based on facial and behavioral approaches. Emotional, behavioral, and cognitive engagement represent distinct dimensions of the learning experience, yet they are deeply interconnected in real-world educational contexts. In the sections that follow, each type of engagement is addressed individually beginning with its theoretical foundations, followed by the rationale for how it was identified and classified in this review, and concluding with an analysis of the detection methods and environments in which it was studied. The broader set of detection approaches used across all papers, including facial expression analysis, gaze tracking, posture estimation, and multimodal methods, is summarized in Fig. 4, which illustrates the prevalence of each approach.

1) EMOTIONAL ENGAGEMENT

Emotional engagement represents the affective aspect of student involvement which emerges through their feelings along with their attitudes and emotional reactions in the learning environment. Students who experience positive emotions of

TABLE 1. Publication venues of the selected studies on student engagement detection using computer vision.

Source	Conference/Journal	References
Conference Paper	Lecture Notes in Computer Science	[35], [36]
	arXiv	[31], [37]
	Artificial Intelligence in Education (AIED)	[38], [39]
	IEEE International Conference on Machine Learning and Applications (ICMLA 2014)	[20], [27]
	IEEE International Conference on Image Processing	[40], [41]
	10th International Conference on Affective Computing & Intelligent Interaction (ACII 2022)	[42]
	2018 International Conference on Multimodal Interaction (ICMI '18)	[43]
	2022 IET International Conference on Engineering Technologies and Applications (IET-ICETA)	[44]
	ACII 2017, Affective Computing and Intelligent Interaction	[45]
	ACM Multimedia 2023	[46]
	ACM SIGITE	[47]
	CCBR 2015 (Lecture Notes in Computer Science)	[48]
	Communications in Computer and Information Science	[49]
	CVPR Workshop 2023	[50]
	European Neural Network Society (EANN 2024)	[51]
	IEEE 18th International Conference on Advanced Learning Technologies (ICALT)	[52]
	IEEE Conference on Digital Avionics Systems (DASC 2019)	[53]
	IEEE Conference on Educational Technologies	[54]
	IEEE Global Engineering Education Conference	[55]
	IEEE Global Engineering Education Conference (EDUCON 2022)	[56]
	IEEE India Conference (INDICON)	[57]
	IEEE International Conference on Advanced Learning Technologies (ICALT)	[58]
	IEEE International Conference on Computing and Networking Technology (ICCNT)	[59]
	IEEE International Conference on Cyber Security and Cloud Computing	[18]
	IEEE International Conference on Engineering and Emerging Technologies	[60]
	IEEE International Conference on Multimodal Interaction	[61]
	IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)	[62]
	IEEE International Conference on Ubiquitous and Future Networks (ICUFN)	[63]
	IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE)	[64]
	IEEE Learning With MOOCs (LWMOOCS) 2020	[65]
	IEEE SmartWorld 2018	[66]
	IEEE Winter Conference on Applications of Computer Vision	[67]
	IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)	[68]
	Immersive Learning Research Network (iLRN)	[69]
	International Bhurban Conference on Applied Sciences & Technology (IBCAST 2023)	[70]
	International Conference on Affective Computing and Intelligent Interaction	[23]
	International Conference on Data Science and Network Security (ICDSNS)	[15]
	International Conference on Innovative Computing and Communications	[16]
	International Conference on Multimodal Interaction	[43]
	2023 IEEE Global Engineering Education Conference (EDUCON)	[71]
	International Conference on Multimodal Interaction (ICMI)	[72]
	Machine Learning and Knowledge Discovery in Databases (ECML PKDD 2019)	[29]
	Sage Science Review of Educational Technology (SSRET) 2022	[73]
	Software Development and Technologies for Enhancing Accessibility (DSAI 2022)	[74]
	SPIE Pattern Recognition and Tracking	[75]
	Springer Lecture Notes in Networks and Systems	[76]
	17th International Conference on Computer Supported Education (CSEDU)	[77]
	Springer MIND 2022	[78]
Journal Article	Technology and Innovation in Learning, Teaching and Education	[79]
	IEEE Transactions on Affective Computing	[80], [81], [82]
	IEEE Transactions on Visualization and Computer Graphics	[83], [84]
	Springer	[26], [85]

TABLE 1. (Continued.) Publication venues of the selected studies on student engagement detection using computer vision.

IEEE ACCESS	[30], [86], [87], [88], [89], [90], [91], [92], [93]
Computers and Electrical Engineering	[14], [15]
Multimedia Tools and Applications	[13], [21], [24], [94], [95], [96], [97], [98]
International Journal of Cognitive Computing in Engineering	[99], [100]
MDPI Electronics	[17], [101]
Sustainability	[102], [103]
Applied Sciences (MDPI)	[104]
Discover Computing	[105]
Discover Education	[106]
Electronics	[22]
Future Generation Computer Systems	[107]
IEEE Transactions on Computational Intelligence and AI in Games	[32]
IEEE Transactions on Learning Technologies	[108]
International Journal of Advanced Computer Science and Applications	[109]
International Journal of Electrical and Electronics Engineering	[110]
International Journal of Information and Education Technology	[111]
International Journal of Web-Based Learning and Teaching Technologies	[112]
Journal of Healthcare Informatics Research	[113]
Journal of Theoretical and Applied Information Technology	[19]
Optical Memory and Neural Networks	[114]
Scalable Computing: Practice and Experience	[47]
SoftwareX	[115]
User Modeling and User-Adapted Interaction	[116]
Applied Mathematics and Nonlinear Sciences	[117], [118]
EURASIP Journal on Image and Video Processing	[119]
IJ. Information Technology and Computer Science	[120]
Indonesian Journal of Electrical Engineering and Computer Science	[121]
International Journal of Intelligent Engineering and Systems	[122]
Journal of Robotics, Networking, and Artificial Life	[123]
Learning and Collaboration Technologies. Human and Technology Ecosystems (HCII 2020)	[124]
Tech Science Press (CMES Journal)	[125]
2025 22nd International Learning and Technology Conference (L&T)	[126]

interest along with enjoyment while feeling satisfied will strengthen their motivation for achievement yet students who experience negative emotions of boredom or anxiety or frustration tend to disengage from their studies and perform poorly.

Emotions also serve as real-time feedback on a student's interaction with the learning material, helping instructors gauge whether the content is resonating or causing confusion.

The fundamental role that emotions play in learning has made facial expressions the primary research method for computer vision systems that evaluate emotional engagement. Extensive work in both affective computing and psychology built on Ekman's research [126], [127] on facial action units created dependable correlations between specific facial muscle shifts and emotional states such as happiness and both anger and confusion. Consequently, facial expression analysis systems have been widely adopted as proxies for emotional engagement in the reviewed studies.

This review identified 52 papers (46%) (e.g. [8], [94], [100], [121]) focusing on emotional engagement, all employing facial expression analysis, as shown in Fig. 3

and 4. The prevalence of this approach is further explained by the dominance of online learning settings, representing 65.49% of studies (e.g. [60], [65], [67], [77]) as shown in see Fig. 5.

Remote learning setups normally reveal faces as the main observable data point through webcams which makes facial expression analysis a natural method to detect student emotions. Facial data-based technologies face specific challenges when used for this purpose. These automated systems face performance limitations because of cultural expression patterns alongside camera measurement variability and student-controlled emotional displays. Hence, their effectiveness requires multiple data processing methods. Computer vision researchers establish facial expression analysis as the leading approach for detecting emotions because of its simple setup combined with obvious facial signals detection capabilities.

2) COGNITIVE ENGAGEMENT

Cognitive engagement reflects the depth of students' mental involvement, capturing their attention, effort, and persistence

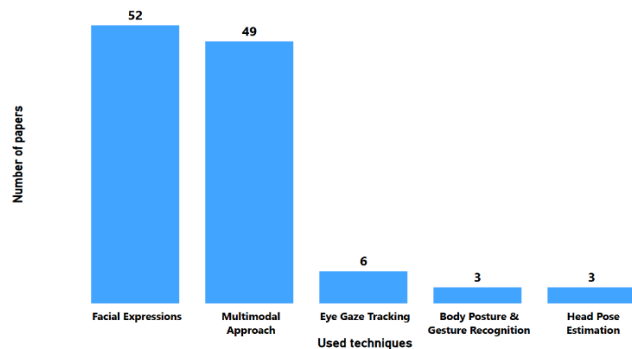


FIGURE 4. Distribution of engagement assessment techniques, showing that facial expressions and multimodal approaches are the most commonly used across reviewed studies.

in understanding complex material. Unlike emotional or behavioral engagement, cognitive engagement is largely internal, making it particularly challenging to detect through visual cues alone. Nevertheless, gaze behavior and head orientation have emerged as viable proxies for attention and mental focus, drawing from research in human-computer interaction and educational psychology.

In this review, gaze tracking and head pose estimation were mapped to cognitive engagement, as they provide indirect visual indicators of students' attention levels. Sustained gaze on learning content is often interpreted as attentional focus, whereas frequent gaze shifts or downward head movements may signal distraction or cognitive fatigue.

Cognitive engagement was investigated in 13.89% of the studies (16 papers), as shown in Fig. 3.

These studies primarily employed gaze tracking (e.g., [16], [17], [55], [56], [62], [68], [82]) and head pose estimation (e.g., [35], [48]) as shown in Fig. 4, often in combination with facial expression analysis in multimodal systems.

However, the inherent limitations of computer vision-based methods in capturing deeper cognitive processes likely explain the relatively lower number of studies focusing solely on this dimension.

Attention-related cues, while useful, remain surface-level indicators that cannot fully encapsulate students' mental effort, highlighting the potential need for integrating non-visual data sources (e.g., task performance logs or physiological sensors) in future research.

3) BEHAVIORAL ENGAGEMENT

Behavioral engagement pertains to the outward manifestations of participation and attentiveness, traditionally assessed through classroom observation techniques. This dimension captures physical actions like sitting upright, raising hands, or interacting with peers, serving as visible markers of a student's involvement with the learning process.

Sustained attention, minimal distractions, and active participation are often interpreted as indicators of behavioral engagement, while slouching, restlessness, or head-nodding may signal disengagement.

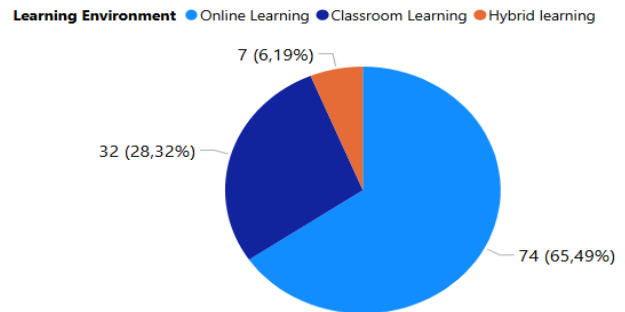


FIGURE 5. Distribution of learning environments in the reviewed studies, showing a strong preference for online learning over classroom and hybrid settings.

In computer vision-based research, posture estimation and gesture recognition systems were mapped to behavioral engagement, as these methods focus on analyzing students' physical behavior. Techniques such as skeletal tracking using OpenPose enable the detection of postural shifts, hand movements, and other bodily cues reflective of attentiveness or fatigue.

Despite its importance in classroom settings, behavioral engagement remains underexplored in the reviewed corpus, accounting for only 2.78% of the papers (e.g., [95], [106], [118]), as shown in Fig. 3.

All these studies employed posture and gesture recognition methods shown in Fig. 4. This limited attention can largely be attributed to the dominance of online learning research (see Fig. 5), where only the upper body is typically visible, reducing the feasibility of full-body behavior analysis. Consequently, behavioral cues that are often better suited to physical classroom environments (e.g., [54], [80], [90], [102], [119]) have received comparatively less focus in computer vision-based engagement detection research.

4) MULTIDIMENSIONAL APPROACHES

Recognizing that engagement is rarely confined to a single dimension, multidimensional approaches seek to capture emotional, cognitive, and behavioral engagement concurrently. These methods reflect the growing consensus that engagement is best assessed through a fusion of affective, attentional, and physical indicators.

Papers adopting multimodal techniques were mapped to this category, as they integrated multiple visual cues to assess engagement holistically. For example, facial expressions were often combined with gaze tracking to infer both emotional states and attention, or paired with posture analysis to link affective responses with physical behavior.

Multidimensional approaches accounted for 38.05% of the studies (43 papers), as shown in Fig. 3. These papers predominantly employed multimodal techniques, with common combinations. The most common pairing was facial expressions with body posture and gesture recognition (9 studies, e.g., [24], [31], [40]), followed by facial expressions with eye gaze tracking (8 studies, e.g., [52], [65], [124]). Other notable

combinations included facial expressions, eye gaze tracking, and head pose estimation (7 studies, e.g., [78], [87], [110]), and facial expressions with head pose estimation (7 studies, e.g., [46], [61], [74]). Additional combinations such as eye gaze tracking with head pose estimation (5 studies, e.g., [18], [19], [42]), or facial expressions with behavioral features, drowsiness detection, or eye blinking were also observed in fewer cases (1–4 studies each).

The analysis highlights that student engagement is a multifaceted construct, with emotional engagement receiving the most attention, followed by growing interest in multidimensional approaches, while behavioral and cognitive engagement remain less explored. Facial expression analysis dominates the field, particularly in online settings, while multimodal techniques are increasingly valued for capturing the complexity of engagement. These methods, however, are inherently tied to the data they rely on, making it crucial to examine the datasets, and features underpinning engagement detection systems, which is the focus of the following section.

B. DATASETS, AND FEATURES FOR ENGAGEMENT DETECTION (RQ2)

Student engagement detection through computer vision represents a critical research field that advanced due to the growing adoption of AI-based learning analytics in educational settings. Engagement represents a multi-dimensional construct that researchers infer by observing behavioral indicators such as facial expressions, gaze direction, head movements and changes in posture. The training and evaluation datasets used to develop detection models directly affect their ability to illustrate real-world learning performance because they establish the models' generalization strength and operational effectiveness. Researchers mostly use camera-based video recording methods to detect engagement because these devices are already present throughout learning environments through devices like laptops and mobile phones along with classroom cameras. The availability of cameras as an engagement-detector makes them the primary data source since they offer inexpensive mass deployment potential without requiring specialized sensing equipment such as eye trackers or physiological sensors.

The availability of appropriate datasets remains a significant obstacle for researchers conducting studies about engagement detection. While public datasets are commonly used to train and evaluate models, most of them were not originally created for engagement detection but rather for emotion recognition or other facial analysis tasks. As a result, researchers often need to adapt and relabel existing datasets to fit engagement-related objectives. In contrast, some studies develop custom datasets, ensuring that the data is collected in real educational settings with appropriate engagement-specific annotations.

This section provides a detailed examination of public datasets, assessing their prevalence, labeling approaches, and relevance to educational research.

1) PUBLIC DATASETS FOR ENGAGEMENT DETECTION

Public datasets serve as a benchmark for engagement detection research, offering standardized resources for model development and comparison.

Table 2 presents a consolidated overview of public datasets used in engagement detection, detailing their main purpose, data environment, type (image/video), labeling approach, and dataset size category.

Dataset Purpose and Relevance to Engagement Detection

Multiple datasets adopted for research about engagement came from projects designed primarily for emotion recognition instead of genuine engagement evaluation. Datasets such as FER-2013, CK+, and RAF-DB were designed to classify emotions like happiness, anger, and surprise, making them valuable for detecting affective states. The concept of engagement goes further than facial expressions because it consists of comprehensive elements that transcend emotional manifestations. The assessment of engagement goes beyond facial emotions because it encompasses sustained attention and cognitive involvement and task-related focus which cannot be solely captured through facial emotion recognition. The low availability of datasets designed for engagement detection leads researchers to adopt emotion datasets for this purpose. The engagement detection datasets DAiSEE, EngageME and the HBCU Engagement Dataset represent some of the only cases designed specifically for this purpose. The datasets include direct annotations which connect to engagement levels rather than using emotions as an intermediate step during processing. Real-world engagement detection tasks require distinct data processing due to the fundamental difference between emotional states and engagement behaviors.

Learning Environments in Public Datasets

Another fundamental limitation of public datasets is that most were collected outside of learning environments. The context in which data is recorded significantly impacts its applicability to education-based engagement detection. Table 1 shows that datasets fall into three primary categories: those collected in controlled lab settings, those obtained in the wild, and a small subset recorded in real learning environments.

Popular datasets including CK+ as well as JAFFE and DISFA+ only obtained their data through precise laboratory settings in which research participants displayed standardized facial expressions. The high-quality data labels from such datasets do not represent the genuine free-form student engagement patterns which naturally emerge in educational environments. The FER-2013 and RAF-DB datasets consist of various real-world captured images but education-specific scenarios are absent from their content. Public datasets that stem from real learning environments compose a minimal section of existing collections. The DAiSEE and EngageME and GESCAM datasets are some of the rare collections which specifically label student engagement behavior within digital learning environments. The lack of learning-specific datasets

creates barriers for direct application of public datasets in classroom engagement detection thus forcing researchers to reanalyze emotional cues for engagement signals or build exclusive datasets to establish this connection.

Labeling Strategies in Public Datasets

A closer look at Table 2 also reveals considerable variation in how engagement is labeled across public datasets. Due to the complexity and subjectivity of engagement particularly in its emotional and behavioral forms, there is no single standardized method for annotating engagement levels, leading to inconsistencies in how datasets are structured.

Public datasets demonstrate substantial differences in the way they characterize educational engagement during annotation processes according to Table 2. There exists no standard annotation approach for multi-dimensional engagement levels because this would create inconsistencies between how different datasets are organized. Most public datasets employ labeling strategies that either divide engagement into multiple fine details or categorize it into fewer multiclass categories. Research studies mainly focus on detecting engagement behaviors as a gradual measure instead of classifying them as either present or absent. Multiple facial expressions receive distinct labels in datasets such as FER-2013 and CK+ and RAF-DB making them applicable for engagement detection tasks. Both DAiSEE and EmotiW employ multiclass labeling which divides engagement into three distinct levels consisting of low, medium and high categories.

In contrast, binary engagement classification remains relatively uncommon. While datasets like MES, SMIC, and UPNA Head Pose provide simple engaged vs. not engaged labels, this approach is often seen as too simplistic for modeling real student behaviors. Engagement is rarely a fixed state, and binary classifications risk overlooking transitional engagement phases, where students fluctuate between focus and distraction.

One particularly rare approach is continuous engagement labeling, where engagement is rated on a dynamic scale rather than discrete categories. EASE is the only dataset in Table 2 that follows this methodology, using a scale from -100 (disengaged) to $+100$ (engaged). The near absence of continuous engagement datasets highlights a gap in current engagement research most studies rely on discrete labels rather than treating engagement as a fluid, evolving state.

Dataset Modality and Size Considerations

Beyond labeling strategies, the modality and scale of public datasets also play a critical role in engagement detection research. Table 2 categorizes datasets based on whether they contain image-based or video-based data, revealing an important trend: while image datasets are more abundant, video datasets provide richer engagement cues.

Image-based datasets like FER-2013, RAF-DB, and CK+ tend to be large-scale, often containing tens of thousands of samples. These datasets are particularly useful for training deep learning models that require vast amounts of data. However, engagement is a time-dependent process, meaning

that single-frame facial expressions may not be sufficient to capture temporal engagement shifts.

Video-based datasets, such as DAiSEE, EngageME, and EmotiW, offer a more comprehensive view of student engagement, as they allow models to analyze how engagement evolves over time. The trade-off, however, is that video-based datasets tend to be computationally demanding, requiring greater resources for model training and inference. Additionally, most engagement-focused video datasets belong to the extra-large category (201+ videos), making them ideal for deep learning applications but less accessible for researchers with limited computational resources.

2) CUSTOM DATASETS FOR ENGAGEMENT DETECTION

While public datasets provide a valuable resource for engagement detection research, their limited relevance to real-world learning environments has led many researchers to develop custom datasets. Unlike public datasets, which are often repurposed from emotion recognition studies, custom datasets are explicitly designed to capture student engagement in authentic educational settings.

A major advantage of custom datasets is that they allow for context-specific data collection, ensuring that engagement is measured in a way that is directly applicable to the learning process. This includes collecting data from classrooms, online courses, and VR-based learning, as well as defining engagement labels based on domain-specific criteria rather than general facial expressions.

Collection Environments in Custom Datasets

The collection environment plays a crucial role in engagement detection, as engagement behaviors can vary significantly across different learning modalities. As shown in Fig. 6, most custom datasets are collected in online learning settings (32 studies), followed by physical classroom environments (21 studies, e.g., [54], [83], [86]). A smaller number of datasets (9 studies) were collected in controlled lab environments, while only one study explored engagement in a VR-based learning setup [68].

These trends reflect the growing importance of online education, where engagement detection is particularly relevant for monitoring student participation in digital courses. However, the fact that lab-based datasets are still used suggests that some studies prefer controlled experimental conditions to ensure data quality, even if it means lower ecological validity.

Annotation Methods in Custom Datasets

One of the key challenges in engagement detection research is how engagement is labeled, because engagement reflects both internal cognitive states and external behaviors, defining annotation methods in custom datasets presents unique challenges. Unlike public datasets, where labels are often predefined (e.g., emotions), custom datasets allow researchers to define their own annotation strategies based on their study objectives.

As shown in Fig. 7, the majority of custom datasets (33 studies) rely on manual annotation, where human observers

TABLE 2. Overview of commonly used datasets, including their application focus, data characteristics, label structure, and number of studies utilizing each.

Datasets Name	Dataset main purpose	Enviroment	Type	number of labels	Original labels	Size Category	Number of studies
FER-2013	Facial Expression FER	In the Wild	Image	Fine-grained (6+)	Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral	Large (10,001-50,000 images)	17
DAiSEE	Facial Expression ED	Learning Environment	Video	Multiclass (3-5)	0: Not Engaged, 1: Barely Engaged, 2: Engaged, 3: Highly Engaged	Extra-Large (201+ videos)	13
EmotiW	Facial Expression ED	In the Wild	Video	Multiclass (3-5)	Disengaged, Low Engagement, Medium Engagement, High Engagement	Large (51-200 videos)	7
CK+	Facial Expression FER	Controlled/Lab	Video	Fine-grained (6+)	Anger, Disgust, Fear, Happiness, Sadness, Surprise, Neutral	Extra-Large (201+ videos)	6
RAF-DB	Facial Expression FER	In the Wild	Image	Fine-grained (6+)	Happy, Sad, Angry, Neutral, Surprise, Fear, Disgust	Large (10,001-50,000 images)	4
CASME II	Facial Expression FER	Controlled/Lab	Video	Multiclass (3-5)	Happiness, Disgust, Surprise, Repression, Others	Extra-Large (201+ videos)	2
JAFPE	Facial Expression FER	Controlled/Lab	Image	Fine-grained (6+)	Happiness, Anger, Sadness, Fear, Surprise, Disgust, Neutral	Small (0-1,000 images)	2
Karolinska Directed Emotional Faces (KDEF)	Facial Expression FER	Controlled/Lab	Image	Fine-grained (6+)	Happiness, Sadness, Surprise, Fear, Disgust, Anger, Neutral	Medium (1,001-10,000 images)	2
OL-SFED	Facial Expression FER	In the Wild	Image	Multiclass (3-5)	Confusion, Distraction, Enjoyment, Neutrality, Fatigue	Large (10,001-50,000 images)	2
RAVDESS	Facial Expression ED	Controlled/Lab	Video	Fine-grained (6+)	Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, Surprised	Extra-Large (201+ videos)	2
SAMM	Facial Expression FER	Controlled/Lab	Video	Fine-grained (6+)	Anger, Contempt, Disgust, Fear, Happiness, Sadness, Surprise	Large (51-200 videos)	2
SMIC	Facial Expression FER	Controlled/Lab	Video	Multiclass (3-5)	Positive, Negative, Surprise	Large (51-200 videos)	2
UPNA Head Pose	Head Pose Estimation	Learning Environment	Video	Binary	Engaged, Not Engaged	Large (51-200 videos)	2
VGAF	Facial Expression FER	In the Wild	Video	Multiclass (3-5)	Positive, Neutral, Negative	Extra-Large (201+ videos)	2
AFEW	Facial Expression FER	In the Wild	Video	Fine-grained (6+)	Angry, Disgust, Fear, Happiness, Sadness, Surprise, Neutral	Extra-Large (201+ videos)	1
BNU-CVWS	Facial Expression ED	Controlled/Lab	Video	Binary	Attentive, Not Attentive	Extra-Large (201+ videos)	1
BNU-LSVED	Facial Expression FER	Controlled/Lab	Video	Multiclass (3-5)	Joviality, Surprise, Concentration, Confusion, Fatigue	Large (51-200 videos)	1
Closed Eyes in the Wild (CEW)	Eye Gaze Tracking	In the Wild	Image	Binary	Open Eye, Closed Eye	Medium (1,001-10,000 images)	1
CMOSE	Facial Expression ED	Controlled/Lab Eye Behavior & Gaze Analysis	Video	Multiclass (3-5)	Highly Disengaged (HD), Disengaged (DE), Engaged (EG), Highly Engaged (HE)	Extra-Large (201+ videos)	1
EASE	Head Pose Estimation	Analysis	Video	Continuous	'-100 (Disengaged) to +100 (Engaged)	Extra-Large (201+ videos)	1
EngageME	Facial Expression ED	Learning Environment	Video	Multiclass (3-5)	Highly Attentive, Moderately Attentive, Low Attention, Inattention	Extra-Large (201+ videos)	1

TABLE 2. (Continued.) Overview of commonly used datasets, including their application focus, data characteristics, label structure, and number of studies utilizing each.

EngageNet	Facial Expression ED	Learning Environment	Video	Multiclass (3-5)	Not-Engaged, Barely-Engaged, Engaged, Highly-Engaged	Extra-Large (201+ videos)	1
Extended DISFA (DISFA+)	Facial Expression FER	Controlled/Lab	Video	Fine-grained (6+)	Sad, Surprise, Anger, Happiness, Fear, Disgust, Happy, Sadness	Extra-Large (201+ videos)	1
GESCAM	Eye Gaze Tracking	Learning Environment	Image	Multiclass (3-5)	Looking at Blackboard, Looking at Notebook, Looking at Mobile Phone	Extra-Large (50,001+ images)	1
HBCU Engagement Dataset	Facial Expression ED	Learning Environment	Video	Multiclass (3-5)	Engaged, Disengaged, Neutral	Extra-Large (201+ videos)	1
Kaggle Eye Gaze Dataset	Eye Gaze Tracking	Eye Behavior & Gaze Analysis	Image	Binary	Engaged, Disengaged	Large (10,001-50,000 images)	1
MRL Eye Dataset	Eye Gaze Tracking	Eye Behavior & Gaze Analysis	Image	Binary	Open Eye, Closed Eye	Extra-Large (50,001+ images)	1
OAHEGA EMOTION RECOGNITION DATASET	Facial Expression FER	Controlled/Lab	Image	Fine-grained (6+)	Ahegao, Contempt, Fear, Angry, Happy, Disgust, Sad, Neutral, Surprise	Extra-Large (50,001+ images)	1
StuEmo24	Facial Expression ED	Learning Environment	Image	Multiclass (3-5)	Engaged, Confused, Bored, Frustrated, Neutral Anger, Contempt, Disgust, Fear, Happiness, Neutral, Sadness, Surprise	Small (0-1,000 images)	1
Yale Face	Face Recognition	Controlled/Lab	Image	Fine-grained (6+)		Small (0-1,000 images)	1

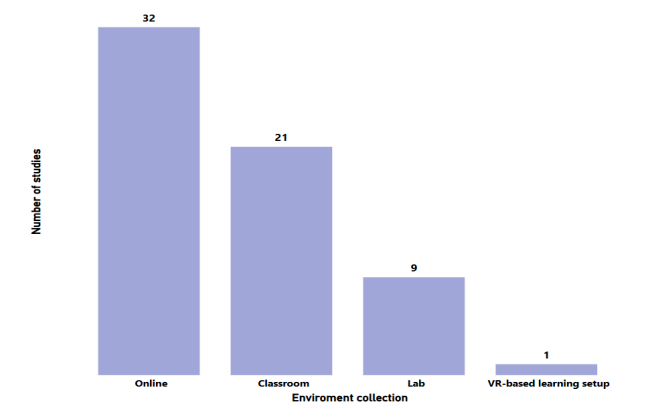


FIGURE 6. Breakdown of dataset collection environments, where online contexts dominate over physical and immersive (VR) setups.

(e.g., teachers, researchers) label engagement levels based on predefined criteria. Hybrid approaches (manual + automatic methods) are also common (19 studies), suggesting that researchers are leveraging automated tools while still incorporating human expertise.

In contrast, self-reported engagement labels (6 studies) are much less frequent, as students may not always accurately assess their own engagement levels. Fully automated annotations (2 studies) and crowdsourced labeling (1 study) remain rare, likely due to the complexity of engagement assessment compared to tasks like emotion recognition.

Engagement Labels in Custom Datasets

Engagement detection models rely on clearly defined engagement levels to train and evaluate machine learning

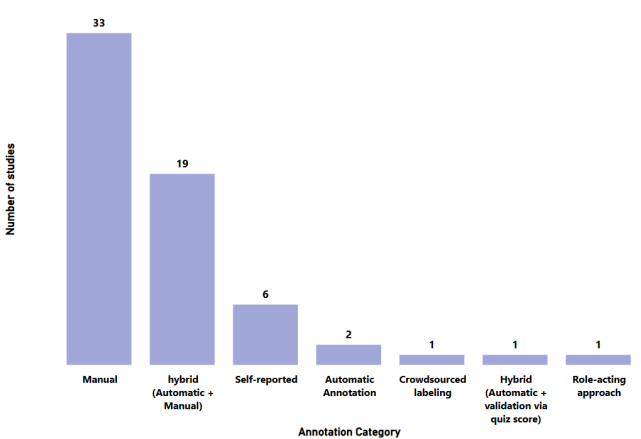


FIGURE 7. Overview of engagement annotation categories in custom datasets, reflecting a clear preference for manual and semi-automated labeling techniques.

algorithms. Fig. 8 illustrates the distribution of engagement labels in custom datasets, revealing a strong preference for multiclass classification (46.03%), where engagement is categorized into low, medium, and high levels.

Binary engagement classification (38.1%) is also common, particularly in studies where the goal is to distinguish between engaged and disengaged states in a simplified manner. However, fine-grained labeling (14.29%), which provides more nuanced engagement states, is less frequently used, indicating that most studies still prefer coarser engagement labels.

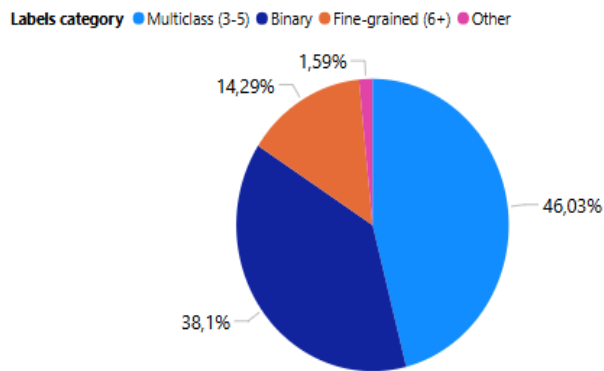


FIGURE 8. Labeling categories used for engagement annotation, reflecting a clear preference for multiclass and binary approaches.

A small proportion of datasets (1.59%) fall into the “Other” category, where engagement is assessed using alternative approaches such as continuous scoring or task-related performance indicators. In these cases, engagement is not labeled into fixed categories but instead tracked as a dynamic variable over time or inferred from student behavior during learning activities. These methods aim to reflect engagement more naturally but remain less common due to their complexity and context dependence.

Dataset Size and Duration Another important aspect of custom datasets is their size (number of participants) and duration of recorded sessions, as these factors impact the robustness and generalizability of engagement detection models.

Fig. 9 shows that most datasets record medium-duration sessions (5-60 min, 60.32%), as engagement tends to fluctuate over this time span. Short-duration studies (<5 min) account for only 15.87%, while long sessions (>60 min) remain relatively rare (9.52%), possibly due to data collection challenges over extended periods.

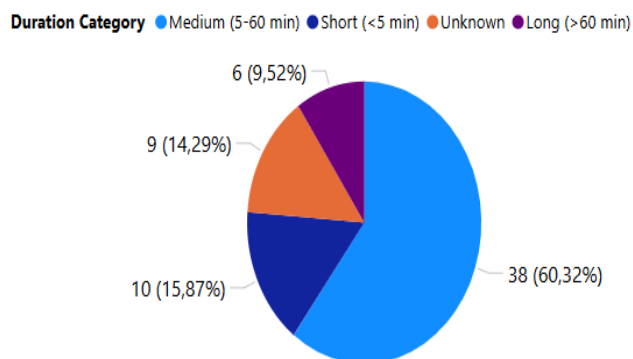


FIGURE 9. Video duration categories in custom datasets, with medium-length recordings (5–60 minutes) being the most prevalent across studies.

In terms of dataset size, Fig. 10 shows that medium-sized datasets (20-100 participants) are the most common

(36.51%), while large-scale datasets (>100 participants) are much less frequent (14.29%). The relatively small size of most custom datasets suggests that engagement studies still face challenges in collecting large-scale, annotated engagement data, possibly due to the labor-intensive nature of engagement labeling.

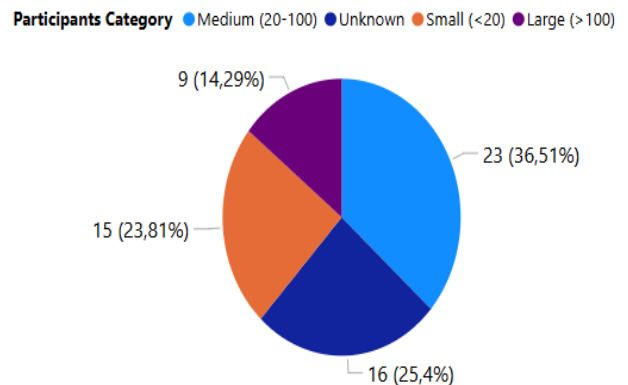


FIGURE 10. Participant size categories used in custom datasets, with most studies involving 20 to 100 participants.

3) FEATURES AND REGIONS OF INTEREST FOR ENGAGEMENT DETECTION

The performance of student engagement detection models depends on both visual information processing methods and the selection of video data features. Learner's engagement detection through computer vision relies on visual elements such as facial expressions, together with eye movements and head posture, and body language features as observational indicators of engagement. The machine learning models utilize these features when processing input data to determine student engagement through analyses of their displayed behaviors.

The following part details the fundamental features detection approach from facial expressions to eye processes and head-body position along with combined multiple detectable features.

Facial Expression Features

People commonly use facial expressions to monitor engagement because such expressions reveal extensive data about emotional states and mental focus levels. Numerous studies of engagement detection depend on Facial Action Units (AUs) taken from the Facial Action Coding System (FACS) to measure facial muscle movements that correspond to various expressions. A person displaying a combination of raised eyebrows (AU1 and AU2) indicates attentiveness, whereas lip compression (AU24) shows frustration. Researchers use emotion recognition models that categorize happy, surprised or neutral expressions for detecting engagement levels but these methods struggle because emotions fail to consistently match the definition of engagement. The commonly used method extracts facial landmarks to locate key face sections including eyes and eyebrows with nose

along with mouth for measuring engagement-related facial motions.

As seen in Table 3, 50 studies rely on facial expressions as the primary engagement feature, making it the most frequently used approach in engagement detection. However, relying solely on facial expressions may overlook critical cognitive and behavioral cues, limiting the completeness of engagement detection models and may not always provide a complete picture of student engagement.

Eye-related Features (Gaze & Blink Patterns)

Eye movements play a crucial role in engagement detection, as they provide strong indicators of visual attention and cognitive load. Gaze direction is one of the most commonly used features, as it tracks whether students are looking at the screen, down at notes, or away from the learning material. Engaged students typically maintain a steady gaze toward the instructional content, while frequent gaze shifts can indicate distraction or disengagement. Blink rate is another widely used feature, as studies suggest that blink frequency changes with cognitive load low engagement or fatigue often results in higher blink rates, whereas focused students blink less frequently. Some studies also explore pupil dilation as an engagement indicator, as pupil size can increase with higher cognitive effort, although this feature is less commonly used in purely computer vision-based models due to its sensitivity to lighting conditions.

As Table 3 highlights, eye-related features are often combined with other engagement cues rather than used alone. While 8 studies use gaze tracking independently, at least 10 studies combine it with facial expressions, and others integrate gaze with head pose estimation (12 studies) to refine engagement classification. This suggests that gaze alone may not be sufficient for engagement detection, but when combined with facial expressions or head movements, it provides a more comprehensive assessment of student attention.

Head & Posture-based Features

Student engagement is not only reflected in facial expressions and eye gaze but also in head movements and body posture, which are particularly useful in both online and physical classroom settings. Head pose estimation is widely used to detect head orientation, including tilting, nodding, and turning, as these movements can indicate whether a student is paying attention or looking away. For instance, frequent head tilts away from the screen may signal distraction, whereas nodding might be a sign of active engagement or agreement with the material.

Beyond head movement, some studies include body posture analysis, as postural cues can provide valuable insights into engagement levels. Research has shown that students who lean forward are often more engaged, while those who slouch or lean backward may exhibit signs of boredom or disengagement. Although body posture alone is rarely used as an engagement predictor, it is often integrated with facial expressions and gaze tracking to enhance engagement assessment.

As seen in Table 3, head pose estimation is widely used, appearing in at least 12 studies in combination with other features, while body posture and gesture recognition appear in 12 studies alongside facial expressions and gaze tracking. While less commonly studied than facial expressions, body posture remains an important complementary feature, particularly in classroom-based engagement detection systems where students' physical activity and movement can provide additional cues beyond facial expressions alone.

Multimodal Feature Combinations

While facial expressions are the most frequently used feature, many engagement detection models integrate multiple features to improve accuracy. Table 3 shows that multimodal approaches are common, with many studies incorporating combinations of facial expressions, gaze tracking, head movements, and body posture analysis. At least 12 studies use facial expressions, gaze tracking, and head pose estimation together, demonstrating that engagement is best captured through a combination of visual cues rather than a single feature alone. Another 12 studies integrate facial expressions with body posture and gesture recognition, particularly in classroom settings where students' physical behaviors provide additional indicators of attentiveness.

Advanced feature combinations include drowsiness detection, eye blinking analysis, and gesture recognition, which can further refine engagement classification. However, the frequent reliance on multimodal features highlights that no single visual cue is sufficient for accurately assessing engagement, as students may display engagement in different ways. Some students may remain still but highly engaged, while others may express engagement through facial expressions and movement. By combining multiple features, engagement detection models can better adapt to the diverse ways students express engagement.

C. PREPROCESSING AND FEATURE ENGINEERING TECHNIQUES (RQ3)

Student engagement detection models function effectively through deep learning systems yet they require proper preparation and structure of raw visual data for successful analysis. The detection of engagement differs from standard computer vision work because it demands interpretation of mental states through minimal outside signals including facial signals and gaze movements with posture information. Real classroom visual data presents multiple analysis challenges to detection systems because of the specific difficulties found in these scenarios.

Publicly accessible engagement detection datasets which serve this purpose do not have designs intended for such applications. These datasets were originally designed to detect emotions and estimate head positions and perform actions but not for engagement measurement. These datasets need detailed reannotation and modification work before they can be suitable for the study of engagement-related behaviors. Real educational settings generate custom datasets which

TABLE 3. Summary of engagement detection approaches based on regions of interest and associated techniques, showing frequency across studies.

Region of Interest	Used techniques detailed	Number of studies
Body Posture	Body Posture & Gesture Recognition	3
Eyes	Eye Gaze Tracking	8
Eyes, Head	(Eye Gaze Tracking + Head Pose Estimation)	5
Face	Facial Expressions	50
Face, Body	(Facial Expressions + Body Posture & Gesture Recognition)	12
Face, eye, head pose	(Facial Expressions + Eye Gaze Tracking + Body Posture & Gesture Recognition)	7
Face, eyes	(Facial Expressions + Eye Gaze Tracking + Drowsiness Detection)	3
Face, eyes	(Facial Expressions + Eye Gaze Tracking)	10
Face, Eyes, head pose	(Facial Expressions + Eye Blinking + Head Pose Estimation)	3
Face, Eyes, head pose	(Facial Expressions + Eye Gaze Tracking + Head Pose Estimation)	12
Face, Eyes, head pose	(Facial Expressions + Head Pose Estimation + Drowsiness Detection)	2
Face, head pose	(Facial Expressions + Head Pose Estimation)	7
head pose	Head Pose Estimation	3

accurately portray engagement dynamics though they face two significant problems because of their restricted diversity with small sample size leading to overfitting.

To address these challenges, preprocessing techniques are applied to clean, normalize, and enhance data quality, while feature engineering ensures that engagement-relevant patterns are extracted in a meaningful way. Together, these steps form the foundation of reliable engagement detection models, bridging the gap between raw visual data and high-level engagement inferences.

1) PREPROCESSING TECHNIQUES FOR ENGAGEMENT DETECTION

Cleaning and Normalization

Visual data used for detecting student engagement usually contains various issues including background noise along with framing and lighting inconsistencies together with resolution problems. The problems in data quality mainly appear in public datasets because these datasets usually record information in uncontrolled environments. The detection models usually get misled due to camera angle variations together with facial position adjustments and unpredictable classroom environment conditions.

To ensure consistency, preprocessing begins with face and pose detection, which isolates the student's facial region or body posture from the background. Techniques such as MTCNN, Haar Cascade, YOLO, OpenPose, and MediaPipe are commonly used to extract facial keypoints or skeletal structures, ensuring that engagement-related cues are the focus of analysis. This step is particularly important in datasets containing multiple individuals, as it prevents models from being influenced by irrelevant background activity.

Lighting and color inconsistencies are addressed through histogram equalization and color-space normalization, which standardize contrast levels and enhance facial feature visibility. Similarly, frame selection techniques are applied in video-based engagement detection to filter out redundant frames, ensuring that models analyze only the most relevant time segments.

For public datasets, reannotation is often necessary before they can be effectively used for engagement detection. Many existing datasets label images for basic emotions or gaze direction, but engagement is a more complex and context-dependent construct. Therefore, researchers often need to manually relabel engagement states or apply semi-supervised learning techniques to refine existing annotations.

Data Augmentation

One of the major limitations of engagement detection datasets is insufficient variability, particularly in custom datasets collected in specific educational environments. When training deep learning models, small datasets often lead to overfitting, where the model learns patterns that are too specific to a particular dataset and fails to generalize to new settings. To address this issue, data augmentation techniques are widely employed to artificially expand dataset diversity.

Geometric transformations such as rotation, scaling, and perspective shifts simulate natural variations in head orientation and posture, ensuring that models generalize well to students with different movement patterns. Noise injection, including Gaussian noise and motion blur, is used to improve robustness by making models less sensitive to minor variations in image quality. More advanced techniques, such as Generative Adversarial Networks (GANs), have been explored to synthesize new engagement-related facial expressions and postures, though their application remains limited in real-world classroom settings.

While augmentation is effective in increasing dataset diversity, careful application is required. Unlike generic image classification tasks, engagement detection relies on subtle behavioral cues, meaning that excessive transformations could distort meaningful patterns. For instance, flipping an image horizontally may not significantly affect object detection, but in engagement detection, it could alter gaze direction or misrepresent head orientation, leading to incorrect classifications.

Handling Imbalanced Engagement Data

Another challenge in engagement detection is class imbalance; certain engagement states, such as high engagement, may be more common than others, leading models to become

biased toward frequently occurring behaviors. This issue is particularly evident in public datasets, where the distribution of engagement levels may not accurately reflect real classroom conditions.

To mitigate this, researchers employ oversampling and undersampling techniques. Methods like SMOTE (Synthetic Minority Over-sampling Technique) create synthetic engagement samples for underrepresented classes, ensuring a more balanced distribution. Alternatively, undersampling reduces the number of dominant-class samples to prevent models from over-relying on majority-class patterns.

Another effective strategy is class-weighted loss functions, which assign higher penalties for misclassified engagement states that are underrepresented, forcing models to pay more attention to less frequent engagement levels. Focal loss functions further refine this approach by dynamically adjusting the importance of misclassified examples, ensuring that models are trained to recognize both common and rare engagement patterns.

By addressing data imbalance through these techniques, engagement detection models become more robust and capable of recognizing a wider range of student behaviors, improving their applicability across diverse learning environments.

2) FEATURE ENGINEERING FOR ENGAGEMENT DETECTION

Engagement detection models rely on structured feature representations that transform raw visual input into meaningful engagement indicators. The reviewed studies employ various feature extraction techniques, ranging from deep learning-based embeddings to handcrafted statistical features, to enhance the accuracy of engagement classification models. Feature engineering primarily focuses on representation learning, multimodal feature fusion, feature selection, and temporal tracking to ensure that engagement cues are effectively captured and interpreted.

Feature Representation and Transformation

Many studies emphasize the importance of reliable and structured encoding of visual signals. For facial expressions, facial landmark detection and AU-based representations (e.g., via OpenFace) are a common foundation. Studies like [40] and [96] apply these techniques to capture subtle muscle movements that indicate attention or emotional shifts. Similarly, study [66] integrates facial feature maps from deep learning models with traditional AU encoding to enhance representation.

Beyond facial encoding, other studies focus on head pose and gaze transformation. Study [18] uses 3D keypoint tracking to compute attention vectors, while study [116] implements HOG-based motion patterns to monitor directional shifts. Study [90] takes this further, generating gaze heatmaps and applying convolutional layers to highlight dominant visual attention zones. From a more handcrafted angle, study [65] applies local directional pattern descriptors to improve texture recognition under variable lighting conditions.

Among the newer additions, study [16] implements facial geometry-based normalization to standardize input before applying LBP-based expression recognition, while study [78] incorporates landmark trajectories across time to reflect the consistency of focus. These approaches help stabilize feature extraction, especially in unconstrained settings.

Multimodal Feature Fusion Strategies

Recognizing that engagement is rarely expressed through a single channel, many studies turn to multimodal feature fusion. This approach combines visual cues from different sources such as face, gaze, posture into a unified engagement representation. Study [17] blends ResNet50 with CBAM modules and temporal convolution layers to integrate expression, gaze, and motion features. In parallel, study [97] uses EmoFAN, OpenFace, and MediaPipe in combination, layering emotional, facial, and skeletal data to reinforce feature depth.

Study [98] employs traditional face detection with additional temporal smoothing to build more consistent feature maps, while study [26] aligns expression and gaze vectors in a joint embedding space, helping the model differentiate between passive staring and genuine attentiveness. Study [21] adds hand gesture tracking to the fusion, reflecting how some students show engagement physically rather than facially, especially in active learning environments.

These integrations highlight a broader trend: no single modality is consistently dominant. Instead, engagement is a composite state, best understood when signals are fused and interpreted together.

Feature Selection and Dimensionality Reduction

As models incorporate increasingly rich features, they face the challenge of dimensionality. Too many inputs can lead to inefficiency and overfitting. To address this, studies like [71] and [86] use attention-based architectures (e.g., DenseNet-CBAM) to focus only on salient features. Meanwhile, study [119] leverages EfficientNet with squeeze-and-excitation blocks to minimize redundancy.

Statistical approaches are also common. Study [113] combines EfficientNet and MobileNet to reduce the feature footprint without losing performance. Study [14], on the other hand, applies a custom feature selection pipeline combining PCA and information gain analysis to prioritize features linked most strongly to engagement labels. This balance of learned and handcrafted filtering strategies allows models to stay both expressive and efficient.

Feature Robustness and data Augmentation

Although many studies do not explicitly employ augmentation techniques, some apply frame-level augmentation strategies to increase dataset diversity and reduce overfitting. Study [45] applies geometric transformations to simulate real-world variations in facial orientation. Study [71] injects motion blur and Gaussian noise to improve CNN resilience under poor video conditions.

Among newly added works, study [13] explores feature-level augmentation by mirroring gaze vectors and simulating different screen positions to mimic how students

might look at various learning resources. Study [52] balances datasets by oversampling underrepresented engagement states, specifically through feature duplication and jittering in time-series data. While augmentation is not yet widespread, it's clear that enhancing the diversity and quality of features during training can significantly improve model robustness, especially when data is limited.

Temporal Feature Engineering for Long-Term Engagement Tracking

Engagement is not static; it fluctuates over time, requiring feature engineering approaches that capture trends rather than isolated observations. Study [38] applies time-series learning techniques to analyze engagement over extended video frames, allowing models to recognize gradual shifts in attention rather than momentary disengagements. Similarly, Study [124] employs Bi-LSTM-based sequential feature learning, ensuring that engagement is tracked continuously rather than frame-by-frame.

Another strategy, seen in Study [71], involves sliding window feature extraction, where engagement cues are aggregated over rolling time intervals, ensuring that models distinguish between brief focus shifts and sustained disengagement. These approaches improve long-term engagement assessment, enabling models to differentiate between momentary distractions (such as a student briefly looking away) and consistent disengagement (such as repeated inattentiveness over a learning session).

Study [85] introduces an engagement memory unit that weighs recent feature history, helping to distinguish between brief lapses and sustained disengagement. Study [28] builds a context window into their feature pipeline, showing that integrating behavior history directly into the feature space can boost both temporal sensitivity and interpretability.

By integrating temporal awareness into feature extraction, engagement models move beyond static classification, enabling dynamic and adaptive engagement monitoring that better reflects real-world learning behaviors.

D. COMPUTER VISION MODELS AND PERFORMANCE EVALUATION (RQ4)

Multiple computer vision models have been studied for student engagement detection but various studies exhibit different rates of model adoption. Research activities in student engagement detection mostly focus on deep learning models specifically CNNs as shown in Fig. 11. The three CNN-based architectures ResNet, VGGNet and Inception-V3 represent the dominant models in this category since they effectively recover spatial information from facial images while recognizing engagement signals through minimal facial expression modifications. The widespread use of CNN-based models for engagement detection remains dominant but researchers have adopted LSTMs and CNN-LSTM hybrid models to track engagement states through time in video-based systems. Real-time systems that require efficient computation utilize MobileNet as their primary choice for model implementation. Meanwhile, traditional machine learning models, such as

SVM and Random Forest, continue to be explored in certain studies, particularly when working with structured feature sets derived from eye gaze, head pose, or handcrafted facial descriptors.

1) MODELS ACROSS DATASETS

It is essential to understand how these selected models have been applied to different datasets beyond pure model selection considerations. The stacked bar chart shown in Fig. 12 demonstrates the frequency at which different models get used for various datasets. Three datasets namely FER-2013, RAF-DB, and DAISEE lead engagement detection studies because researchers use them most frequently. FER-2013 and RAF-DB have been primarily used with CNN-based architectures, reinforcing their strength in facial expression analysis. DAISEE, a video-based dataset, has seen a mix of CNN and CNN-LSTM applications, reflecting the growing interest in temporal engagement modeling.

Interestingly, models such as ResNet and Inception-V3 are consistently evaluated on multiple datasets, suggesting their adaptability to different engagement contexts. Conversely, some models appear only in isolated studies or within specific datasets, indicating that their suitability might be more constrained to certain engagement detection settings.

The diversity of datasets used across studies highlights the challenge of standardization in engagement detection research. While CNN-based models perform well across multiple datasets, their effectiveness may be dataset-dependent, particularly when engagement cues extend beyond facial expressions to incorporate head movements or posture changes.

2) EVALUATION METRICS FOR ENGAGEMENT DETECTION

An assessment of engagement detection model performance relies on different evaluation metrics yet accuracy stands out as the most popular metric. According to Fig. 13 accuracy stands as the most frequently used evaluation metric appearing in 66.1% of research studies. The limitations of accuracy measurements for unbalanced engagement datasets have prompted researchers to introduce additional evaluation metrics including F1-score at 12.7% and Mean Squared Error with 8.4%. F1-score shows great value in situations with imbalanced class distributions because it offers suitable measurement for engagement levels that do not match dataset distributions.

Engagement detection takes a regression form when researchers use Mean Squared Error (MSE) instead of classification methods given its ability to work with continuous engagement measurements.

Several studies have utilized AUC-ROC while also employing mAP and correlation-based metrics but these evaluation measures have not received widespread adoption. When working with datasets having multiple behavioral cues that overlap, F1-score should be considered over accuracy because it offers enhanced evaluation precision.

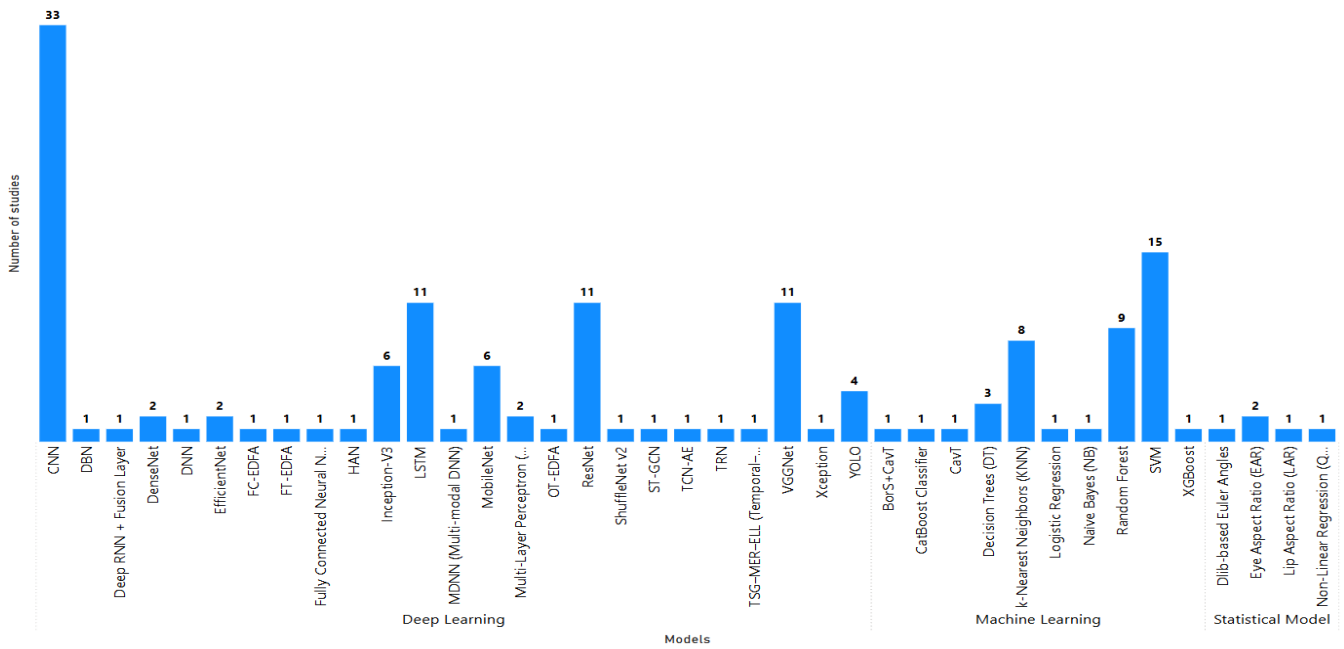


FIGURE 11. Distribution of student engagement detection algorithms used across the selected studies, with CNN, VGGNet, LSTM, and SVM emerging as the most frequently implemented models.

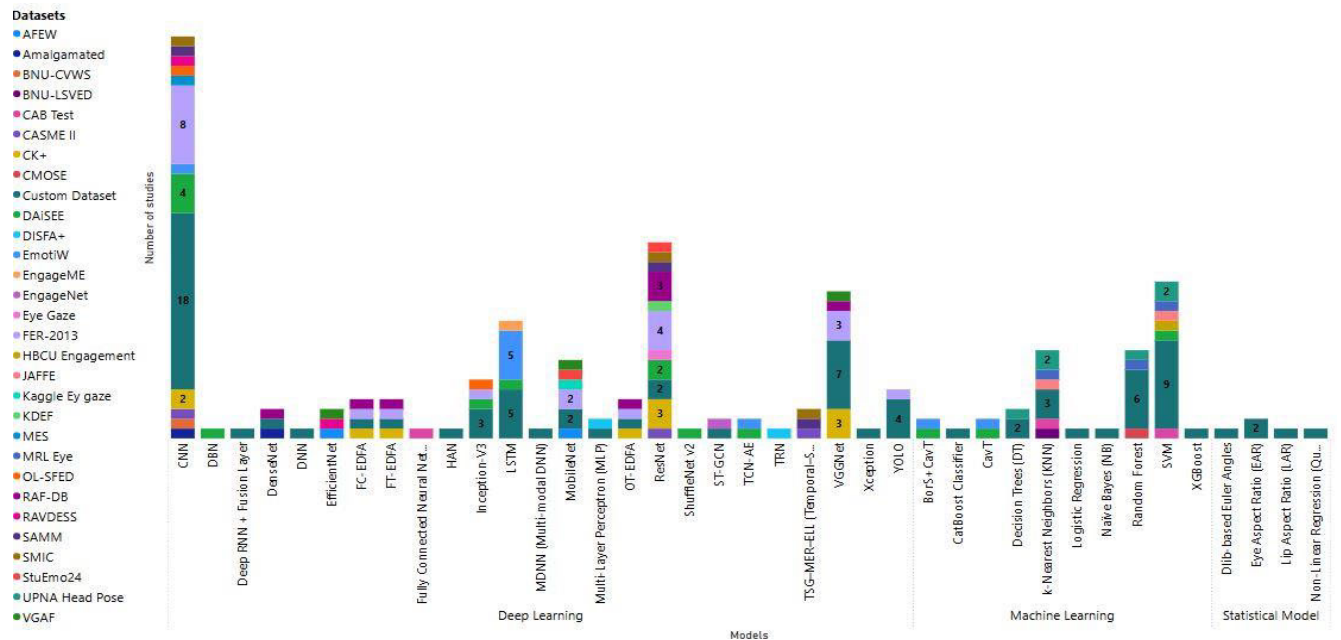


FIGURE 12. Mapping of engagement detection algorithms to the datasets used in selected studies, illustrating the widespread use of CNNs across multiple datasets and highlighting model-dataset diversity in deep learning and machine learning approaches.

The overreliance on accuracy suggests a potential area for improvement, encouraging future studies to adopt a more diverse set of evaluation metrics for a comprehensive assessment of engagement detection models. Fig. 14 shows the Number of studies per metrics.

Given the diversity of datasets, it is essential to assess how models perform across different data sources. The box plot

(Fig. 14) illustrates the distribution of performance across public datasets, highlighting significant variations in reported accuracy. Several key observations emerge:

FER-2013 and RAF-DB consistently yield high performance across models, likely due to their well-structured annotations and their alignment with facial emotion recognition tasks, where CNNs excel.

Performance Metrics • Accuracy • Others • Mean Squared Error (MSE) • F1 Score • AUC • mAP

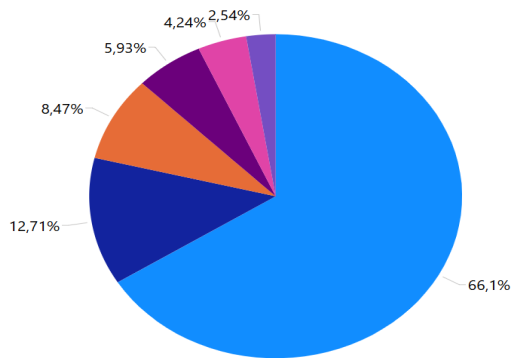


FIGURE 13. Distribution of performance metrics reported in selected studies, indicating a strong preference for accuracy-based evaluation.

DAISEE exhibits a broader range of performance scores, reflecting the complexity of engagement detection in video-based settings. The variability suggests that engagement analysis in dynamic learning environments remains a challenging task, particularly when models must capture attention shifts over time.

Datasets incorporating head pose and gaze tracking (e.g., UPNA Head Pose, EngageNet) show moderate accuracy levels, reinforcing the idea that beyond facial expressions, engagement cues derived from non-facial behaviors introduce additional modeling complexity.

These findings emphasize that dataset characteristics heavily influence model performance, and comparisons between studies must consider these variations. The high accuracy reported on facial expression-based datasets should not be generalized to all engagement detection tasks, particularly those incorporating multi-modal engagement indicators such as gaze and head movements.

3) MODEL PERFORMANCE ACROSS DIFFERENT APPROACHES

Beyond dataset-related variability, model selection also plays a critical role in engagement detection performance. The box plot in 1 compares the performance of different models, distinguishing between deep learning and machine learning approaches.

CNN-based models consistently achieve high accuracy, with relatively low variance across studies. This supports their robustness in static image-based engagement detection.

LSTM-based models, while promising for video analysis, exhibit greater performance fluctuations, indicating that temporal engagement modeling remains an area for further refinement.

SVM, Random Forest, and Decision Trees show a wider spread in performance, reinforcing that their effectiveness is dataset-dependent. While some studies report high accuracy, others highlight limitations in handling the complexity of engagement cues.

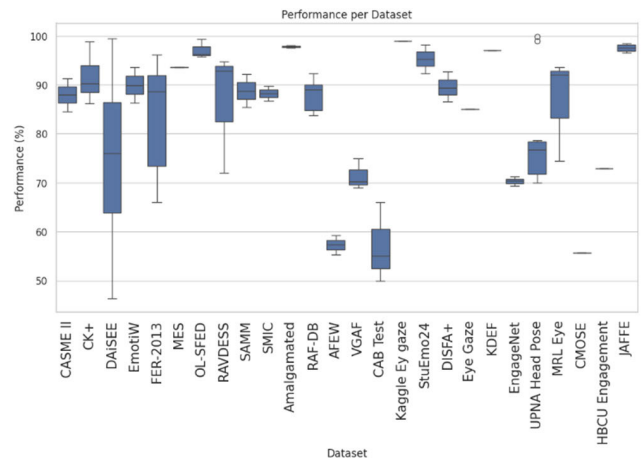


FIGURE 14. Performance distribution of engagement detection models across datasets, highlighting variability in model effectiveness depending on dataset characteristics.

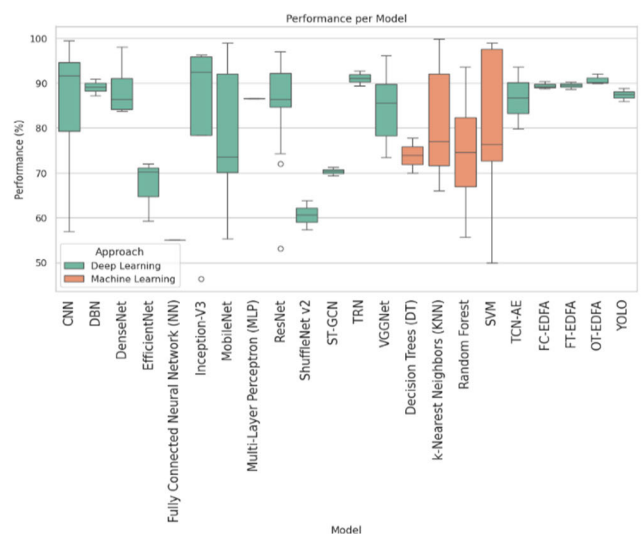


FIGURE 15. Box plot illustrating the performance variation of engagement detection models, categorized by deep learning and machine learning techniques.

Part of this variability can also be attributed to differences in preprocessing strategies and feature engineering techniques. Traditional models such as SVM and Random Forest often depend on carefully crafted features, including facial landmarks, head pose estimations, or gaze vectors. In these cases, performance is closely tied to the quality and consistency of the preprocessing pipeline. Even in deep learning approaches, data augmentation, normalization, and temporal smoothing can influence accuracy, particularly in video-based studies where frame-level fluctuations may affect model stability. Therefore, performance comparisons must be contextualized within the broader modeling workflow, including not only the architecture itself but also how the input data is prepared.

Overall, these results align with previous findings on dataset-dependent model performance, emphasizing the

importance of choosing the right model for a given engagement detection task.

4) PERFORMANCE ON PRIVATE DATASETS

While public datasets provide a standardized foundation for benchmarking, a substantial number of studies also rely on private datasets, particularly those collected in real classroom or remote learning environments. These datasets are often tailored to specific research questions, contexts, or demographic groups, offering richer contextual information but lower generalizability.

Performance results on private datasets tend to vary more widely than on public datasets. In some cases, models trained and evaluated on private data report high accuracies occasionally exceeding 90%, which may reflect the controlled conditions or tailored model-dataset alignment. However, these results should be interpreted with caution. The lack of transparency and standardized benchmarks in private datasets can lead to challenges in reproducibility and comparative evaluation.

Notably, CNN-based models still dominate in private dataset applications, but there is also a greater presence of traditional machine learning approaches, especially when the dataset is small or when engagement labels are annotated through observation or surveys rather than automation. In these scenarios, SVM and Random Forest have demonstrated reasonable performance, particularly when paired with feature engineering techniques that reduce noise and emphasize relevant visual cues.

It is important to consider that performance reported on private datasets, while often promising, may not translate directly to other learning environments. Thus, future research should aim to bridge the gap between private and public dataset findings

V. DISCUSSION

This systematic literature review investigated the landscape of student engagement detection using computer vision, focusing on detection types, feature modalities, dataset use, and model performance. The findings provide several critical insights while also revealing challenges and

Significant progress has been made in the field of engagement detection, particularly in recognizing emotional states through facial analysis, a modality that has long dominated both computational and educational engagement research. Prior reviews, such as those focused on sentiment analysis in online learning [11], have further reinforced this emphasis by analyzing emotion as a key proxy for engagement. This modality has become the cornerstone for many systems due to its scalability and ease of integration into online learning environments. However, as the field matures, expanding the focus toward behavioral and cognitive dimensions offers the potential to enrich models and make them more representative of the diverse ways students express engagement.

This review identified four primary engagement types: emotional, behavioral, cognitive, and multidimensional.

While emotional engagement remains the most extensively studied, our findings show growing interest in cognitive engagement, often inferred through gaze and head orientation, which provide insight into mental focus. Still, these proxies remain limited in distinguishing deeper learning states such as reflection or critical thinking. Behavioral engagement, typically captured through posture or physical interaction, is especially underrepresented particularly in online settings, where body cues are often truncated or absent. As previously suggested by [10], this lack of behavioral data limits the capacity of models to holistically assess engagement. Our review confirms and expands this concern by emphasizing how contextual constraints like camera placement or hardware limitations further restrict behavioral cue integration.

In contrast to earlier SLRs that focused primarily on traditional engagement measures (e.g., surveys, learning logs, sentiment analysis), this review highlights a clear shift toward multimodal systems, where researchers combine facial expressions, gaze, head pose, and posture to build more comprehensive engagement profiles. Although direct comparisons between fusion strategies remain limited, we observed diverse implementation of both early fusion (feature-level integration) and late fusion (decision-level combination). Early fusion supports richer joint representations but struggles with synchronization across modalities. Late fusion, while more modular and robust to missing data, can lose nuanced correlations between cues. Few studies explicitly benchmark these strategies; future work should explore standardized fusion benchmarks to evaluate trade-offs in complexity, latency, and interpretability.

A notable development is the increased feasibility of real-time systems, especially with the adoption of lightweight architectures such as MobileNet, YOLOv5, and quantized CNNs. While computational constraints remain a concern, especially for deployment in resource-constrained settings (e.g., rural schools or mobile devices), current results suggest that scalable, real-time engagement detection is within reach. However, latency-performance trade-offs are rarely reported consistently, indicating a need for clearer metrics when designing for practical classroom integration.

Another emerging trend is the transition from static snapshots of engagement to temporal modeling, reflecting growing interest in how engagement fluctuates over time. Recurrent and sequential models (LSTM, GRU) and transformer-based approaches are being increasingly used to capture these dynamic affective trajectories, which align with recent theoretical work on engagement as a temporal construct. Unlike traditional static classification, these methods model shifts, peaks, and drops in engagement offering a more realistic and actionable perspective for educators and adaptive systems. Prior reviews [12] have touched on temporal dynamics, but our review shows that such approaches are now gaining more technical traction, with multiple studies successfully implementing sequence-aware models to track engagement episodes over minutes or entire sessions.

The application of engagement detection in immersive environments, including VR, AR, and the Metaverse, represents one of the most exciting expansions of the field. Head-mounted displays, eye-tracking sensors, and spatial tracking offer an unprecedented array of visual and behavioral signals that can be leveraged to infer engagement. Although still in its infancy, this domain has been formally explored in studies like Safari Bazargani et al. (2021, 2025) [128], [129], which offer early frameworks for understanding engagement in metaverse-based learning. Complementing this, Huang et al. (2021) [130] conducted a systematic review of AR and VR in language learning and found that immersive environments significantly enhance motivation, interaction, and reduce learning anxiety all of which are strongly correlated with student engagement. These findings suggest that visual cues traditionally used in engagement detection (e.g., facial expressions, eye gaze, body posture) not only remain relevant in immersive settings but may become even more informative when captured through VR/AR hardware. Our taxonomy supports these extensions by showing that modality-context combinations (e.g., body posture in VR or eye movement in AR) are now both feasible and necessary for next-generation systems. As educational platforms adopt immersive modalities, engagement detection must adapt accordingly to remain relevant and effective.

Equally important is the ethical landscape surrounding engagement detection. Video-based monitoring raises significant privacy and surveillance concerns, especially in classroom settings where student consent and autonomy may be limited. Additionally, cultural and neurodiversity factors affect how engagement is expressed and interpreted what looks like disengagement in one culture or context may be attentiveness in another. These concerns call for careful attention to bias mitigation, dataset diversity, and explainable AI. Encouragingly, some recent studies have begun embedding participatory design and fairness audits into their systems, reflecting a positive shift from purely performance-driven models to responsible AI frameworks.

To provide a clearer view of the field's structure and help identify underexplored areas, this review proposes a high-level taxonomy structured along three key axes:

- Modality: face, eyes, posture, gestures, voice
- Context: in-person, online, VR/AR
- Engagement Type: emotional, behavioral, cognitive

This taxonomy not only organizes the current landscape but invites future studies to systematically report which axes they operate on. For example, few studies explore behavioral engagement in immersive settings a combination that may offer unique advantages but is currently underrepresented.

This review followed a systematic protocol, but several limitations should be acknowledged. First, the search was restricted to English-language, peer-reviewed publications from 2014 to 2025, potentially excluding relevant non-English or gray literature. Second, the review focused exclusively on computer vision-based engagement detection, omitting multimodal systems where vision is not the primary

modality (e.g., speech-only systems). Third, despite efforts to apply objective screening criteria, subjectivity in interpretation and classification is an inherent risk in qualitative synthesis. Finally, while we included a broad range of studies across educational contexts, rapid advancements in AI architectures may mean some very recent contributions were not captured within the review window.

The importance of engagement detection extends far beyond model accuracy or architecture design. Low student engagement is a leading predictor of dropout, especially in online and higher education. Early detection systems can enable timely intervention, helping instructors adjust strategies or provide personalized support before students disengage entirely. This is particularly critical for asynchronous and remote learners, who may lack real-time instructor feedback.

There are also important social equity implications. Engagement detection systems can play a role in supporting neurodiverse learners, identifying disengagement patterns in students who may not signal it in traditional ways. Additionally, economically marginalized or underrepresented groups may benefit from adaptive learning systems that can dynamically adjust content based on observed engagement. In the long term, improving student engagement contributes to higher retention, better academic outcomes, and more inclusive education systems. These educational impacts translate to economic gains for individuals and societies through improved employability, reduced dropout, and increased productivity.

To advance the field, several key research avenues should be prioritized:

- Deepening research on behavioral and cognitive engagement detection, particularly in real-world classrooms
- Developing efficient, real-time engagement models deployable across a range of hardware platforms
- Benchmarking fusion strategies (early vs. late) and establishing temporal modeling standards
- Expanding engagement detection into VR/AR/metaverse contexts with new multimodal pipelines
- Embedding ethical, inclusive, and privacy-preserving principles throughout system design and evaluation

By aligning technical innovation with social responsibility and educational priorities, the field of engagement detection is well-positioned to shape the next generation of intelligent, adaptive learning systems.

VI. CONCLUSION

In this systematic literature review, we examined the landscape of student engagement detection using computer vision techniques, highlighting key trends, challenges, and opportunities. Our findings indicate that emotional engagement has been the most extensively studied dimension, largely driven by the widespread use of facial expression analysis. At the same time, we observed a significant rise in multimodal approaches, which combine visual cues such as facial expressions, gaze behavior, and body posture to provide a more

comprehensive understanding of student engagement. These approaches reflect a growing awareness of the complex, multidimensional nature of engagement, though they often require more sophisticated models and richer datasets. In the other hand, cognitive and behavioral dimensions remain underexplored.

We found that most existing datasets used in engagement detection were originally designed for emotion recognition or facial analysis tasks, limiting their relevance for educational settings. Custom datasets, while more context-specific, are typically small and lack standardization, making cross-study comparisons difficult. This highlights a pressing need for publicly available, large-scale datasets specifically developed for engagement detection across diverse learning environments.

We observed that deep learning models, especially CNNs and CNN-LSTM hybrids, dominate the field due to their ability to capture spatial and temporal engagement cues. However, we also noted a promising shift toward multimodal approaches, which integrate facial, gaze, and posture data to provide a more comprehensive understanding of engagement. These methods, while effective, require robust preprocessing pipelines, large-scale annotated datasets, and greater computational resources, challenges that researchers are actively working to address.

Through this review, we emphasize the importance of developing standardized, diverse, and education-specific datasets to support the next generation of engagement detection models. We also advocate for the advancement of explainable and lightweight AI systems that can operate in real-time, particularly in online and resource-constrained learning environments. Looking ahead, several directions offer fertile ground for continued development:

- Deepening research on behavioral and cognitive engagement detection
- Designing efficient, real-time models for diverse hardware settings
- Establishing comparative benchmarks for fusion strategies and temporal modeling
- Expanding engagement systems into immersive digital environments
- Embedding ethical considerations throughout the design pipeline

Taken together, these areas reflect a dynamic and maturing research landscape. By aligning technical advances with social values and educational needs, the field is well-positioned to shape the next generation of intelligent learning systems.

APPENDIX

The dataset supporting this systematic literature review including the list of included studies, metadata, and summaries has been submitted to Mendeley Data and is currently under moderation. Once approved, it will be publicly available at: <https://doi.org/10.17632/bm2rm98764.1>

REFERENCES

- [1] K. Schnitzler, D. Holzberger, and T. Seidel, "All better than being disengaged: Student engagement patterns and their relations to academic self-concept and achievement," *Eur. J. Psychol. Educ.*, vol. 36, no. 3, pp. 627–652, Sep. 2021, doi: [10.1007/s10212-020-00500-6](https://doi.org/10.1007/s10212-020-00500-6).
- [2] A. F. Mena-Guacas, L. López-Catalán, C. Bernal-Bravo, and C. Ballesteros-Regaña, "Educational transformation through emerging technologies: Critical review of scientific impact on learning," *Educ. Sci.*, vol. 15, no. 3, p. 368, Mar. 2025, doi: [10.3390/educsci15030368](https://doi.org/10.3390/educsci15030368).
- [3] S. Maloney, M. Axelsen, L. Galligan, J. Turner, P. Redmond, A. Brown, M. Basson, and J. Lawrence, "Using LMS log data to explore student engagement with coursework videos," *Online Learn.*, vol. 26, no. 4, pp. 399–423, Dec. 2022, doi: [10.24059/olj.v26i4.2998](https://doi.org/10.24059/olj.v26i4.2998).
- [4] M. A. A. Dewan, M. Murshed, and F. Lin, "Engagement detection in online learning: A review," *Smart Learn. Environments*, vol. 6, no. 1, p. 1, Dec. 2019, doi: [10.1186/s40561-018-0080-z](https://doi.org/10.1186/s40561-018-0080-z).
- [5] L. Mishra, T. Gupta, and A. Shree, "Online teaching-learning in higher education during lockdown period of COVID-19 pandemic," *Int. J. Educ. Res. Open*, vol. 1, Jan. 2020, Art. no. 100012, doi: [10.1016/j.ijedro.2020.100012](https://doi.org/10.1016/j.ijedro.2020.100012).
- [6] H. A. Poonja, M. A. Shirazi, M. J. Khan, and K. Javed, "Engagement detection and enhancement for STEM education through computer vision, augmented reality, and haptics," *Image Vis. Comput.*, vol. 136, Aug. 2023, Art. no. 104720, doi: [10.1016/j.imavis.2023.104720](https://doi.org/10.1016/j.imavis.2023.104720).
- [7] J. A. Fredricks, P. C. Blumenfeld, and A. H. Paris, "School engagement: Potential of the concept, state of the evidence," *Rev. Educ. Res.*, vol. 74, no. 1, pp. 59–109, Mar. 2004, doi: [10.3102/00346543074001059](https://doi.org/10.3102/00346543074001059).
- [8] P. Buono, B. De Carolis, F. D'Errico, N. Macchiarulo, and G. Palestra, "Assessing student engagement from facial behavior in on-line learning," *Multimedia Tools Appl.*, vol. 82, no. 9, pp. 12859–12877, Apr. 2023, doi: [10.1007/s11042-022-14048-8](https://doi.org/10.1007/s11042-022-14048-8).
- [9] G. Lamé, "Systematic literature reviews: An introduction," in *Proc. Design Soc. Int. Conf. Eng. Design*, Jul. 2019, vol. 1, no. 1, pp. 1633–1642, doi: [10.1017/dsi.2019.169](https://doi.org/10.1017/dsi.2019.169).
- [10] L. Subramanian and M. A. Mahmoud, "A systematic review on students' engagement in classroom: Indicators, challenges and computational techniques," *Int. J. Adv. Comput. Sci. Appl.*, vol. 11, no. 1, 2020, doi: [10.14569/ijacsa.2020.0110113](https://doi.org/10.14569/ijacsa.2020.0110113).
- [11] A. Anwar, I. U. Rehman, M. M. Nasralla, S. B. A. Khattak, and N. Khilji, "Emotions matter: A systematic review and meta-analysis of the detection and classification of Students' emotions in STEM during online learning," *Educ. Sci.*, vol. 13, no. 9, p. 914, Sep. 2023, doi: [10.3390/educsci13090914](https://doi.org/10.3390/educsci13090914).
- [12] M. Ilić, V. Mikić, L. Kopanja, and B. Vesin, "Intelligent techniques in e-learning: A literature review," *Artif. Intell. Rev.*, vol. 56, no. 12, pp. 14907–14953, Dec. 2023, doi: [10.1007/s10462-023-10508-1](https://doi.org/10.1007/s10462-023-10508-1).
- [13] S. Gupta, P. Kumar, and R. K. Tekchandani, "Facial emotion recognition based real-time learner engagement detection system in online learning context using deep learning models," *Multimedia Tools Appl.*, vol. 82, no. 8, pp. 11365–11394, Mar. 2023, doi: [10.1007/s11042-022-13558-9](https://doi.org/10.1007/s11042-022-13558-9).
- [14] P. Bhardwaj, P. K. Gupta, H. Panwar, M. K. Siddiqui, R. Morales-Menendez, and A. Bhaik, "Application of deep learning on student engagement in e-learning environments," *Comput. Electr. Eng.*, vol. 93, Jul. 2021, Art. no. 107277, doi: [10.1016/j.compeleceng.2021.107277](https://doi.org/10.1016/j.compeleceng.2021.107277).
- [15] M. Lima, K. R. Ahmed, N. Jahan, I. Mahmud, M. S. Parvez, and P. Revathi, "Deep learning based approach for detecting student engagement through facial emotions," in *Proc. Int. Conf. Data Sci. Netw. Secur. (ICDSNS)*, Jul. 2024, pp. 1–6, doi: [10.1109/ICDSNS62112.2024.10691098](https://doi.org/10.1109/ICDSNS62112.2024.10691098).
- [16] Z. A. T. Ahmed, M. E. Jadhav, A. M. Al-madani, M. Tawfik, S. N. Alsubari, and A. A. A. Shareef, "Real-time detection of student engagement: Deep learning-based system," in *Proc. Int. Conf. Innov. Comput. Commun.* Singapore: Springer, 2021, pp. 313–323, doi: [10.1007/978-981-16-2594-7_26](https://doi.org/10.1007/978-981-16-2594-7_26).
- [17] M. U. Uçar and E. Özdemir, "Recognizing students and detecting student engagement with real-time image processing," *Electronics*, vol. 11, no. 9, p. 1500, May 2022, doi: [10.3390/electronics11091500](https://doi.org/10.3390/electronics11091500).
- [18] Y. Wang, A. Kotha, P.-h. Hong, and M. Qiu, "Automated student engagement monitoring and evaluation during learning in the wild," in *Proc. 7th IEEE Int. Conf. Cyber Secur. Cloud Comput. (CSCloud)/ 6th IEEE Int. Conf. Edge Comput. Scalable Cloud (EdgeCom)*, Aug. 2020, pp. 270–275, doi: [10.1109/CSCloud-EdgeCom49738.2020.00054](https://doi.org/10.1109/CSCloud-EdgeCom49738.2020.00054).

- [19] A. Alrawahneh and D. S. B. Safei, "A model of video watching concentration level measurement among students using head pose and eye tracking detection," *J. Theor. Appl. Inf. Technol.*, vol. 99, no. 17, pp. 4305–4315, Sep. 2021.
- [20] S. Aslan, Z. Cataltepe, I. Diner, O. Dundar, A. A. Esme, R. Ferens, G. Kamhi, E. Oktay, C. Soysal, and M. Yener, "Learner engagement measurement and classification in 1:1 learning," in *Proc. 13th Int. Conf. Mach. Learn. Appl.*, Dec. 2014, pp. 545–552, doi: [10.1109/ICMLA.2014.111](https://doi.org/10.1109/ICMLA.2014.111).
- [21] S. Gupta, P. Kumar, and R. Tekchandani, "A multimodal facial cues based engagement detection system in e-learning context using deep learning approach," *Multimedia Tools Appl.*, vol. 82, no. 18, pp. 28589–28615, Jul. 2023, doi: [10.1007/s11042-023-14392-3](https://doi.org/10.1007/s11042-023-14392-3).
- [22] Y. Qi, L. Zhuang, H. Chen, X. Han, and A. Liang, "Evaluation of students' learning engagement in online classes based on multimodal vision perspective," *Electronics*, vol. 13, no. 1, p. 149, Dec. 2023, doi: [10.3390/electronics13010149](https://doi.org/10.3390/electronics13010149).
- [23] A. Kaur, A. Mustafa, L. Mehta, and A. Dhali, "Prediction and localization of student engagement in the wild," in *Proc. Digit. Image Comput., Techn. Appl. (DICTA)*, Dec. 2018, pp. 1–8, doi: [10.1109/DICTA.2018.8615851](https://doi.org/10.1109/DICTA.2018.8615851).
- [24] C. Chang, C. Zhang, L. Chen, and Y. Liu, "An ensemble model using face and body tracking for engagement detection," in *Proc. 20th ACM Int. Conf. Multimodal Interact.*, Oct. 2018, pp. 616–622, doi: [10.1145/3242969.3264986](https://doi.org/10.1145/3242969.3264986).
- [25] X. Ai, V. S. Sheng, C. Li, and Z. Cui, "Class-attention video transformer for engagement intensity prediction," 2022, *arXiv:2208.07216*.
- [26] D. Boulanger, M. A. A. Dewan, V. S. Kumar, and F. Lin, "Lightweight and interpretable detection of affective engagement for online learners," in *Proc. IEEE Int. Conf. Dependable, Autonomic Secure Comput., Int. Conf. Pervasive Intell. Comput., Int. Conf. Cloud Big Data Comput., Int. Conf. Cyber Sci. Technol. Congr. (DASC/PiCom/CBDCCom/CyberSciTech)*, Oct. 2021, pp. 176–184, doi: [10.1109/DASC-PiCom-CBDCCom-CyberSciTech52372.2021.00040](https://doi.org/10.1109/DASC-PiCom-CBDCCom-CyberSciTech52372.2021.00040).
- [27] A. Harb, A. Gad, M. Yaghi, M. Alhalabi, H. Zia, J. Yousef, A. Khelifi, K. Ghoudi, and M. Ghazal, "Diverse distant-students deep emotion recognition and visualization," *Comput. Electr. Eng.*, vol. 111, Nov. 2023, Art. no. 108963, doi: [10.1016/j.compeleceng.2023.108963](https://doi.org/10.1016/j.compeleceng.2023.108963).
- [28] O. M. Nezami, M. Dras, L. Hamey, D. Richards, S. Wan, and C. Paris, "Automatic recognition of student engagement using deep learning and facial expression," 2018, *arXiv:1808.02324*.
- [29] K. Watanabe, T. Sathyanarayana, A. Dengel, and S. Ishimaru, "EnGauge: Engagement gauge of meeting participants estimated by facial expression and deep neural network," *IEEE Access*, vol. 11, pp. 52886–52898, 2023, doi: [10.1109/ACCESS.2023.3279428](https://doi.org/10.1109/ACCESS.2023.3279428).
- [30] A. Abedi and S. S. Khan, "Engagement measurement based on facial landmarks and spatial-temporal graph convolutional networks," 2024, *arXiv:2403.17175*.
- [31] A. Psaltis, K. C. Apostolakis, K. Dimitropoulos, and P. Daras, "Multimodal student engagement recognition in prosocial games," *IEEE Trans. Games*, vol. 10, no. 3, pp. 292–303, Sep. 2018, doi: [10.1109/TCI-AIG.2017.2743341](https://doi.org/10.1109/TCI-AIG.2017.2743341).
- [32] D. Budgen and P. Brereton, "Performing systematic literature reviews in software engineering," in *Proc. 28th Int. Conf. Softw. Eng.*, May 2006, pp. 1051–1052, doi: [10.1145/1134285.1134500](https://doi.org/10.1145/1134285.1134500).
- [33] M. J. Page, "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. 71, Mar. 2021, doi: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71).
- [34] W. Lee, D. Alessio, W. Rebelsky, S. Satish Gattupalli, H. Yu, I. Arroyo, M. Betke, S. Bargal, T. Murray, F. Sylvia, and B. P. Woolf, "Measurements and interventions to improve student engagement through facial expression recognition," in *Adaptive Instructional Systems* (Lecture Notes in Computer Science), vol. 13332. Cham, Switzerland: Springer, 2022, pp. 286–301, doi: [10.1007/978-3-031-05887-5_20](https://doi.org/10.1007/978-3-031-05887-5_20).
- [35] D. Canedo, A. Trifan, and A. J. R. Neves, "Focus estimation in academic environments using computer vision," in *Pattern Recognition and Image Analysis* (Lecture Notes in Computer Science), vol. 11867. Cham, Switzerland: Springer, 2019, pp. 620–628, doi: [10.1007/978-3-030-31332-6_54](https://doi.org/10.1007/978-3-030-31332-6_54).
- [36] C.-H. Wu, S.-Y. Liu, X. Huang, X. Wang, R. Zhang, L. Minciullo, W. K. Yiu, K. Kwan, and K.-T. Cheng, "CMOSE: Comprehensive multi-modality online student engagement dataset with high-quality labels," 2023, *arXiv:2312.09066*.
- [37] A. Linson, Y. Xu, A. R. English, and R. B. Fisher, "Identifying student struggle by analyzing facial movement during asynchronous video lecture viewing: Towards an automated tool to support instructors," in *Artificial Intelligence in Education* (Lecture Notes in Computer Science), vol. 13355, M. M. Rodrigo, N. Matsuda, A. I. Cristea, and V. Dimitrova, Eds., Cham, Switzerland: Springer, 2022, pp. 53–65.
- [38] S. Yadav, M. N. Siddiqui, Y. Vats, and J. Shukla, "EngageME: Exploring neuropsychological tests for assessing attention in online learning," in *Artificial Intelligence in Education* (Lecture Notes in Computer Science), vol. 14829, A. M. Olney, I.-A. Chounta, Z. Liu, O. C. Santos, and I. I. Bittencourt, Eds., Cham, Switzerland: Springer, 2024, pp. 394–408, doi: [10.1007/978-3-031-64302-6_28](https://doi.org/10.1007/978-3-031-64302-6_28).
- [39] I. Alkabbany, A. Ali, A. Farag, I. Bennett, M. Ghanoum, and A. Farag, "Measuring student engagement level using facial information," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 3337–3341, doi: [10.1109/ICIP.2019.8803590](https://doi.org/10.1109/ICIP.2019.8803590).
- [40] Y.-Y. Li and Y.-P. Hung, "Feature fusion of face and body for engagement intensity detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 3312–3316, doi: [10.1109/ICIP.2019.8803488](https://doi.org/10.1109/ICIP.2019.8803488).
- [41] A. Augusma, "Multimodal perception and statistical modeling of pedagogical classroom events using a privacy-safe non-individual approach," in *Proc. 10th Int. Conf. Affect. Comput. Intell. Interact. Workshops Demos (ACIIW)*, Oct. 2022, pp. 1–5, doi: [10.1109/ACIIW57231.2022.10086029](https://doi.org/10.1109/ACIIW57231.2022.10086029).
- [42] A. Kaur, "Attention network for engagement prediction in the wild," in *Proc. 20th ACM Int. Conf. Multimodal Interact.*, Oct. 2018, pp. 516–519, doi: [10.1145/3242969.3264972](https://doi.org/10.1145/3242969.3264972).
- [43] C.-H. Hsia, B. Chiang, L.-Y. Ke, Z.-Y. Ciou, and C.-F. Lai, "Student engagement analysis using facial expression in online course," in *Proc. IET Int. Conf. Eng. Technol. Appl. (IET-ICETA)*, Oct. 2022, pp. 1–2, doi: [10.1109/IET-ICETA56553.2022.9971565](https://doi.org/10.1109/IET-ICETA56553.2022.9971565).
- [44] B. M. Booth, A. M. Ali, S. S. Narayanan, I. Bennett, and A. A. Farag, "Toward active and unobtrusive engagement assessment of distance learners," in *Proc. 7th Int. Conf. Affect. Comput. Intell. Interact. (ACII)*, Oct. 2017, pp. 470–476.
- [45] H. Yu, D. A. Alessio, W. Lee, W. Rebelsky, F. Sylvia, T. Murray, J. J. Magee, I. Arroyo, B. P. Woolf, S. A. Bargal, and M. Betke, "COVES: A cognitive-affective deep model that personalizes math problem difficulty in real time and improves student engagement with an online tutor," in *Proc. 31st ACM Int. Conf. Multimedia*, Oct. 2023, pp. 6152–6160, doi: [10.1145/3581783.3613965](https://doi.org/10.1145/3581783.3613965).
- [46] Z. Li, J. Li, M. Han, and M. Yang, "A computer-vision based engagement evaluation system for more effective learning design," in *Proc. 22nd Annu. Conf. Inf. Technol. Educ.*, Oct. 2021, pp. 57–58, doi: [10.1145/3450329.3478312](https://doi.org/10.1145/3450329.3478312).
- [47] C. Tang, P. Xu, Z. Luo, G. Zhao, and T. Zou, "Automatic facial expression analysis of students in teaching environments," in *Biometric Recognition* (Lecture Notes in Computer Science), vol. 9428. Cham, Switzerland: Springer, 2015, pp. 439–447, doi: [10.1007/978-3-319-25417-3_52](https://doi.org/10.1007/978-3-319-25417-3_52).
- [48] D. Canedo, A. Trifan, and A. J. R. Neves, "Monitoring students' attention in a classroom through computer vision," in *Highlights of Practical Applications of Agents, Multi-Agent Systems, and Complexity: The PAAMS Collection* (Communications in Computer and Information Science), vol. 887. Cham, Switzerland: Springer, 2018, pp. 371–378, doi: [10.1007/978-3-319-94779-2_32](https://doi.org/10.1007/978-3-319-94779-2_32).
- [49] A. M. Mathew, A. A. Khan, T. Khalid, and R. Souissi, "GESCAM : A dataset and method on gaze estimation for classroom attention measurement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 636–645, doi: [10.1109/CVPRW63382.2024.00068](https://doi.org/10.1109/CVPRW63382.2024.00068).
- [50] A. Sassi, S. Chérif, and W. Jaafar, "Intelligent framework for monitoring student emotions during online learning," in *Engineering Applications of Neural Networks* (Communications in Computer and Information Science), vol. 2141. Cham, Switzerland: Springer, 2024, pp. 207–219, doi: [10.1007/978-3-031-62495-7_16](https://doi.org/10.1007/978-3-031-62495-7_16).
- [51] T. S. Ashwin and R. M. R. Guddeti, "Unobtrusive students' engagement analysis in computer science laboratory using deep learning techniques," in *Proc. IEEE 18th Int. Conf. Adv. Learn. Technol. (ICALT)*, Jul. 2018, pp. 436–440, doi: [10.1109/ICALT.2018.00110](https://doi.org/10.1109/ICALT.2018.00110).

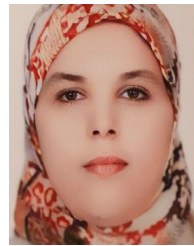
- [52] M. Murshed, M. A. A. Dewan, F. Lin, and D. Wen, "Engagement detection in e-learning environments using convolutional neural networks," in *Proc. IEEE Int. Conf. Dependable, Autonomic Secure Comput., Int. Conf. Pervasive Intell. Comput., Int. Conf. Cloud Big Data Comput., Int. Conf. Cyber Sci. Technol. Congr. (DASC/PiCom/CBDCCom/CyberSciTech)*, Aug. 2019, pp. 80–86, doi: [10.1109/DASC/PiCom/CBDCCom/CYBERSCTECH.2019.00028](https://doi.org/10.1109/DASC/PiCom/CBDCCom/CYBERSCTECH.2019.00028).
- [53] S. Deniz, D. Lee, G. Kurian, L. Altamirano, D. Yee, M. Ferra, B. Hament, J. Zhan, L. Gewali, and P. Oh, "Computer vision for attendance and emotion analysis in school settings," in *Proc. IEEE 9th Annu. Comput. Commun. Workshop Conf. (CCWC)*, Jan. 2019, pp. 0134–0139, doi: [10.1109/CCWC.2019.8666488](https://doi.org/10.1109/CCWC.2019.8666488).
- [54] S. K. Vishnumolakala, V. S. Vallamkonda, N. P. Subheesh, and J. Ali, "In-class student emotion and engagement detection system (iSEEDS): An AI-based approach for responsive teaching," in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, May 2023, pp. 1–5, doi: [10.1109/EDUCON54358.2023.10125254](https://doi.org/10.1109/EDUCON54358.2023.10125254).
- [55] A. R. Khan, S. Khosravi, S. Hussain, R. Ghannam, A. Zoha, and M. A. Imran, "EXECUTE: Exploring eye tracking to support E-learning," in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, Mar. 2022, pp. 670–676, doi: [10.1109/EDUCON52537.2022.9766506](https://doi.org/10.1109/EDUCON52537.2022.9766506).
- [56] R. Gopinathan, N. and E. Sherly, "Visual attention score and fatigue level measure of students through eye analysis-machine learning approach," in *Proc. IEEE 19th India Council Int. Conf. (INDICON)*, Nov. 2022, pp. 1–5, doi: [10.1109/INDICON56171.2022.10040215](https://doi.org/10.1109/INDICON56171.2022.10040215).
- [57] S. Mrayhi, M. K. Khribi, and M. Jemni, "Enhancing MOOCs through real-time learner engagement and emotion detection using computer vision and machine learning," in *Proc. IEEE Int. Conf. Adv. Learn. Technol. (ICALT)*, Jul. 2024, pp. 1–2, doi: [10.1109/ICALT61570.2024.00007](https://doi.org/10.1109/ICALT61570.2024.00007).
- [58] P. Duraisamy, R. S. Gadaley, J. P. van Haneghan, and S. Jackson, "Smaller class room engagement using appropriate seating arrangements," in *Proc. 10th Int. Conf. Comput., Commun. Netw. Technol. (ICCCNT)*, Jul. 2019, pp. 1–7, doi: [10.1109/ICCCNT45670.2019.8944390](https://doi.org/10.1109/ICCCNT45670.2019.8944390).
- [59] S. Tennakoon, T. Wickramaarachchi, R. Weerakotuwa, P. Sulochana, A. Karunasena, and V. Piyawardana, "E-pod: E-learning system for improving student engagement in asynchronous mode," in *Proc. Int. Conf. Eng. Emerg. Technol. (ICEET)*, Oct. 2021, pp. 1–6, doi: [10.1109/ICEET53442.2021.9659592](https://doi.org/10.1109/ICEET53442.2021.9659592).
- [60] D. Chiaro, D. Annuziata, S. Izzo, and F. Piccialli, "Unveiling engagement in virtual classrooms: A multimodal analysis," in *Proc. IEEE Int. Conf. Big Data (BigData)*, Dec. 2023, pp. 4761–4769, doi: [10.1109/BIG-DATA59044.2023.10386484](https://doi.org/10.1109/BIG-DATA59044.2023.10386484).
- [61] S. S. Mane and A. R. Surve, "Engagement detection using video-based estimation of head movement," in *Proc. 3rd IEEE Int. Conf. Recent Trends Electron., Inf. Commun. Technol. (RTEICT)*, May 2018, pp. 1745–1749, doi: [10.1109/RTEICT42901.2018.9012323](https://doi.org/10.1109/RTEICT42901.2018.9012323).
- [62] A. Desai, S. Jain, L. Marodi, and S. V., "An affect computing based attention estimation," in *Proc. 11th Int. Conf. Ubiquitous Future Netw. (ICUFN)*, Jul. 2019, pp. 330–332, doi: [10.1109/ICUFN.2019.8806053](https://doi.org/10.1109/ICUFN.2019.8806053).
- [63] N. S. Nuha, T. Mahmud, N. Rezaaana, M. S. Hossain, and K. Andersson, "An approach of analyzing classroom student engagement in multimodal environment by using deep learning," in *Proc. IEEE 9th Int. Women Eng. (WIE) Conf. Electr. Comput. Eng. (WIECON-ECE)*, Nov. 2023, pp. 286–291, doi: [10.1109/WIECON-ECE60392.2023.10456488](https://doi.org/10.1109/WIECON-ECE60392.2023.10456488).
- [64] P. Duraisamy, J. Van Haneghan, J. Thomas, and R. S. Gadaley, "Online classroom engagement observation using deep learning," in *Proc. IEEE Learn. With MOOCs (LWMOOCs)*, Sep. 2020, pp. 81–83, doi: [10.1109/LWMOOCs50143.2020.9234339](https://doi.org/10.1109/LWMOOCs50143.2020.9234339).
- [65] M. A. A. Dewan, F. Lin, D. Wen, M. Murshed, and Z. Uddin, "A deep learning approach to detecting engagement of online learners," in *Proc. IEEE SmartWorld, Ubiquitous Intell. Comput., Adv. Trusted Comput., Scalable Comput. Commun., Cloud Big Data Comput., Internet People Smart City Innov. (SmartWorld/SCALCOM/UIC/ATC/CBDCCom/IOP/SCI)*, Oct. 2018, pp. 1895–1902, doi: [10.1109/SMARTWORLD.2018.00318](https://doi.org/10.1109/SMARTWORLD.2018.00318).
- [66] S. Dhamija and T. E. Boulton, "Automated action units vs. expert raters: Face off," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2018, pp. 259–268, doi: [10.1109/WACV.2018.00035](https://doi.org/10.1109/WACV.2018.00035).
- [67] T. Lee, D. Kim, S. Park, D. Kim, and S.-J. Lee, "Predicting mind-wandering with facial videos in online lectures," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 2103–2112, doi: [10.1109/CVPRW56347.2022.00228](https://doi.org/10.1109/CVPRW56347.2022.00228).
- [68] T. H. N. Huy, M. K. J. Carlon, and J. S. Cross, "Simulated gaze tracking using computer vision for virtual reality," in *Proc. 8th Int. Conf. Immersive Learn. Res. Netw. (iLRN)*, May 2022, pp. 1–5, doi: [10.23919/iLRN55037.2022.9815891](https://doi.org/10.23919/iLRN55037.2022.9815891).
- [69] N. Siddiqui, M. Khalid, A. Ahmed, A. Aleem, M. Irfan, W. Ahmad, and F. B. Muslim, "An AI-based classroom monitoring system leveraging computer vision and machine learning," in *Proc. 20th Int. Bhurban Conf. Appl. Sci. Technol. (IBCAST)*, Aug. 2023, pp. 273–278, doi: [10.1109/IBCAST59916.2023.10713048](https://doi.org/10.1109/IBCAST59916.2023.10713048).
- [70] M. Irfan, P. Patel, and B. Hassan, "EngageSense: A hybrid approach for real time engagement detection for virtual classrooms," in *Proc. IEEE Global Eng. Educ. Conf. (EDUCON)*, Apr. 2025, pp. 1–9.
- [71] M. Fraile, C. Fawcett, J. Lindblad, N. Sladoje, and G. Castellano, "End-to-end learning and analysis of infant engagement during guided play: Prediction and explainability," in *Proc. Int. Conf. Multimodal Interact.*, Nov. 2022, pp. 444–454, doi: [10.1145/3536221.3556629](https://doi.org/10.1145/3536221.3556629).
- [72] A. S. Pillai, "Student engagement detection in classrooms through computer vision and deep learning: A novel approach using YOLOv4," *Sage Sci. Rev. Educ. Technol.*, vol. 5, no. 1, pp. 87–97, 2022.
- [73] S. R. Khanal, P. Sharma, K. Thapa, H. Fernandes, J. Barroso, and V. Filipe, "Performance analysis and evaluation of cloud vision emotion APIs," in *Proc. 11th Int. Conf. Softw. Develop. Technol. Enhancing Accessibility Fighting Info-Exclusion*, Nov. 2024, pp. 215–221.
- [74] S. L. Naidu, H. Beбина, P. Bhatia, P. Duraisamy, J. V. Haneghan, and T. Sandhan, "Classroom engagement evaluation using 360-degree view of the camera with deep learning techniques," in *Pattern Recognition and Tracking XXXIV*. Bellingham, WA, USA: SPIE, 2023, p. 13, doi: [10.1117/12.2664147](https://doi.org/10.1117/12.2664147).
- [75] P. Ravi and M. Ali Akber Dewan, "Real-time multi-module student engagement detection system," in *Communication and Intelligent Systems (Lecture Notes in Networks and Systems)*, vol. 686. Singapore: Springer, 2023, pp. 261–278.
- [76] A. Ayari, M. Chaabouni, and H. B. Ghezala, "A deep learning approach for automatic detection of learner engagement in educational context," in *Proc. 17th Int. Conf. Comput. Supported Educ.*, 2025, pp. 372–379, doi: [10.5220/0013283200003932](https://doi.org/10.5220/0013283200003932).
- [77] N. Ahmad, A. Gupta, and D. Singh, "Using deep transfer learning to predict student engagement in online courses," in *Machine Learning, Image Processing, Network Security and Data Sciences (Communications in Computer and Information Science)*, vol. 1763, 2022, pp. 27–36, doi: [10.1007/978-3-031-24367-7_3](https://doi.org/10.1007/978-3-031-24367-7_3).
- [78] P. Sharma, S. Joshi, S. Gautam, S. Maharjan, S. R. Khanal, M. C. Reis, J. Barroso, and V. M. de Jesus Filipe, "Student engagement detection using emotion analysis, eye tracking and head movement with machine learning," in *Technology and Innovation in Learning, Teaching and Education (Communications in Computer and Information Science)*, vol. 1720. Cham, Switzerland: Springer, 2022, pp. 52–68, doi: [10.1007/978-3-031-22918-3_5](https://doi.org/10.1007/978-3-031-22918-3_5).
- [79] J. Whitehill, Z. Serpell, Y.-C. Lin, A. Foster, and J. R. Movellan, "The faces of engagement: Automatic recognition of student engagement from facial expressions," *IEEE Trans. Affect. Comput.*, vol. 5, no. 1, pp. 86–98, Jan. 2014, doi: [10.1109/TAFFC.2014.2316163](https://doi.org/10.1109/TAFFC.2014.2316163).
- [80] Ö. Sümer, P. Goldberg, S. D'Mello, P. Gerjets, U. Trautwein, and E. Kasneci, "Multimodal engagement analysis from facial videos in the classroom," *IEEE Trans. Affect. Comput.*, vol. 14, no. 2, pp. 1012–1027, Apr. 2023, doi: [10.1109/TAFFC.2021.3127692](https://doi.org/10.1109/TAFFC.2021.3127692).
- [81] H. Monkaresi, N. Bosch, R. A. Calvo, and S. K. D'Mello, "Automated detection of engagement using video-based estimation of facial expressions and heart rate," *IEEE Trans. Affect. Comput.*, vol. 8, no. 1, pp. 15–28, Jan. 2017, doi: [10.1109/TAFFC.2016.2515084](https://doi.org/10.1109/TAFFC.2016.2515084).
- [82] S. M. Asish, A. K. Kulshreshtha, C. W. Borst, and S. Sutradhar, "Classification of internal and external distractions in an educational VR environment using multimodal features," *IEEE Trans. Vis. Comput. Graphics*, vol. 30, no. 11, pp. 7332–7342, Nov. 2024, doi: [10.1109/TVCG.2024.3456207](https://doi.org/10.1109/TVCG.2024.3456207).
- [83] H. Zeng, X. Shu, Y. Wang, Y. Wang, L. Zhang, T.-C. Pong, and H. Qu, "EmotionCues: Emotion-oriented visual summarization of classroom videos," *IEEE Trans. Vis. Comput. Graphics*, vol. 27, no. 7, pp. 3168–3181, Jul. 2021, doi: [10.1109/TVCG.2019.2963659](https://doi.org/10.1109/TVCG.2019.2963659).
- [84] A. Abedi and S. S. Khan, "Detecting disengagement in virtual learning as an anomaly using temporal convolutional network autoencoder," 2022, *arXiv:2211.06870*.

- [85] R. Kawamura, S. Shirai, N. Takemura, M. Alizadeh, M. Cukurova, H. Takemura, and H. Nagahara, "Detecting drowsy learners at the wheel of e-learning platforms with multimodal learning analytics," *IEEE Access*, vol. 9, pp. 115165–115174, 2021, doi: [10.1109/ACCESS.2021.3104805](https://doi.org/10.1109/ACCESS.2021.3104805).
- [86] L. Li and D. Yao, "Emotion recognition in complex classroom scenes based on improved convolutional block attention module algorithm," *IEEE Access*, vol. 11, pp. 143050–143059, 2023, doi: [10.1109/ACCESS.2023.3340510](https://doi.org/10.1109/ACCESS.2023.3340510).
- [87] A. Sukumaran and A. Manoharan, "Multimodal engagement recognition from image traits using deep learning techniques," *IEEE Access*, vol. 12, pp. 25228–25244, 2024, doi: [10.1109/ACCESS.2024.3353053](https://doi.org/10.1109/ACCESS.2024.3353053).
- [88] U. L. Sowjanya and R. Krithiga, "Decoding student emotions: An advanced CNN approach for behavior analysis application using uniform local binary pattern," *IEEE Access*, vol. 12, pp. 106273–106284, 2024, doi: [10.1109/ACCESS.2024.3436531](https://doi.org/10.1109/ACCESS.2024.3436531).
- [89] J. Pordoy, H. Farman, N. K. Dicheva, A. Anwar, M. M. Nasralla, N. Khilji, and I. U. Rehman, "Multi-frame transfer learning framework for facial emotion recognition in e-learning contexts," *IEEE Access*, vol. 12, pp. 151360–151381, 2024, doi: [10.1109/ACCESS.2024.3478072](https://doi.org/10.1109/ACCESS.2024.3478072).
- [90] T. S. Ashwin and R. M. R. Guddeti, "Unobtrusive behavioral analysis of students in classroom environment using non-verbal cues," *IEEE Access*, vol. 7, pp. 150693–150709, 2019, doi: [10.1109/ACCESS.2019.2947519](https://doi.org/10.1109/ACCESS.2019.2947519).
- [91] A. Sukumaran and A. Manoharan, "Student engagement recognition: Comprehensive analysis through EEG and verification by image traits using deep learning techniques," *IEEE Access*, vol. 13, pp. 11639–11662, 2025, doi: [10.1109/ACCESS.2025.3526187](https://doi.org/10.1109/ACCESS.2025.3526187).
- [92] R. Saleem and M. Aslam, "A multi-faceted deep learning approach for student engagement insights and adaptive content recommendations," *IEEE Access*, vol. 13, pp. 69236–69256, 2025, doi: [10.1109/ACCESS.2025.3561459](https://doi.org/10.1109/ACCESS.2025.3561459).
- [93] S. K. Gupta, T. S. Ashwin, and R. M. R. Guddeti, "Students' affective content analysis in smart classroom environment using deep learning techniques," *Multimedia Tools Appl.*, vol. 78, no. 18, pp. 25321–25348, Sep. 2019, doi: [10.1007/s11042-019-7651-z](https://doi.org/10.1007/s11042-019-7651-z).
- [94] A. Pise, H. Vadapalli, and I. Sanders, "Facial emotion recognition using temporal relational network: An application to E-learning," *Multimedia Tools Appl.*, vol. 81, no. 19, pp. 26633–26653, Aug. 2022, doi: [10.1007/s11042-020-10133-y](https://doi.org/10.1007/s11042-020-10133-y).
- [95] M. Rashmi, T. S. Ashwin, and R. M. R. Guddeti, "Surveillance video analysis for student action recognition and localization inside computer laboratories of a smart campus," *Multimedia Tools Appl.*, vol. 80, no. 2, pp. 2907–2929, Jan. 2021, doi: [10.1007/s11042-020-09741-5](https://doi.org/10.1007/s11042-020-09741-5).
- [96] M. Aly, "Revolutionizing online education: Advanced facial expression recognition for real-time student progress tracking via deep learning model," *Multimedia Tools Appl.*, vol. 84, no. 13, pp. 12575–12614, Jun. 2024, doi: [10.1007/s11042-024-19392-5](https://doi.org/10.1007/s11042-024-19392-5).
- [97] A. Abedi, C. Thomas, D. B. Jayagopi, and S. S. Khan, "Bag of states: A non-sequential approach to video-based engagement measurement," *Multimedia Syst.*, vol. 30, no. 1, Feb. 2024, doi: [10.1007/s00530-023-01244-1](https://doi.org/10.1007/s00530-023-01244-1).
- [98] S. Gupta, P. Kumar, and R. Tekchandani, "EDFA: Ensemble deep CNN for assessing student's cognitive state in adaptive online learning environments," *Int. J. Cognit. Comput. Eng.*, vol. 4, pp. 373–387, Jun. 2023, doi: [10.1016/j.jcce.2023.11.001](https://doi.org/10.1016/j.jcce.2023.11.001).
- [99] B. Abdellaoui, A. Remaida, Z. Sabri, M. Abdellaoui, A. El Hafidy, Y. El Bouzekri El Idriissi, and A. Moumen, "Analyzing emotions in online classes: Unveiling insights through topic modeling, statistical analysis, and random walk techniques," *Int. J. Cognit. Comput. Eng.*, vol. 5, pp. 221–236, Jan. 2024, doi: [10.1016/j.jcce.2024.05.003](https://doi.org/10.1016/j.jcce.2024.05.003).
- [100] H.-J. Lee and D. Lee, "Study of process-focused assessment using an algorithm for facial expression recognition based on a deep neural network model," *Electronics*, vol. 10, no. 1, p. 54, Dec. 2020, doi: [10.3390/electronics10010054](https://doi.org/10.3390/electronics10010054).
- [101] C. Llurba, G. Fretes, and R. Palau, "Classroom emotion monitoring based on image processing," *Sustainability*, vol. 16, no. 2, p. 916, Jan. 2024, doi: [10.3390/su16020916](https://doi.org/10.3390/su16020916).
- [102] S. Han, D. Yusupbag, X. Yao, and Z. Han, "Investigating the emotion patterns of students' abnormal interactions in primary class teaching contexts," *Sustainability*, vol. 15, no. 24, p. 16785, Dec. 2023, doi: [10.3390/su152416785](https://doi.org/10.3390/su152416785).
- [103] Y. Hu, Z. Jiang, and K. Zhu, "An optimized CNN model for engagement recognition in an E-learning environment," *Appl. Sci.*, vol. 12, no. 16, p. 8007, Aug. 2022, doi: [10.3390/app12168007](https://doi.org/10.3390/app12168007).
- [104] H. Zhao, B.-G. Kim, A. Slowik, and D. Pan, "Temporal-spatial correlation and graph attention-guided network for micro-expression recognition in english learning livestreams," *Discover Comput.*, vol. 27, no. 1, p. 47, Nov. 2024, doi: [10.1007/s10791-024-09477-y](https://doi.org/10.1007/s10791-024-09477-y).
- [105] K. Kaewkaisorn, K. Pintong, S. Bunyang, T. Tansawat, and T. Siriborvornratanakul, "Student attentiveness analysis in virtual classroom using distraction, drowsiness and emotion detection," *Discover Educ.*, vol. 3, no. 1, p. 56, May 2024, doi: [10.1007/s44217-024-00117-7](https://doi.org/10.1007/s44217-024-00117-7).
- [106] T. S. Ashwin and R. M. R. Guddeti, "Affective database for e-learning and classroom environments using Indian students' faces, hand gestures and body postures," *Future Gener. Comput. Syst.*, vol. 108, pp. 334–348, Jul. 2020, doi: [10.1016/j.future.2020.02.075](https://doi.org/10.1016/j.future.2020.02.075).
- [107] X. Tian, B. P. Nunes, Y. Liu, and R. Manrique, "Predicting student engagement using sequential ensemble model," *IEEE Trans. Learn. Technol.*, vol. 17, pp. 939–950, 2024, doi: [10.1109/TLT.2023.3342860](https://doi.org/10.1109/TLT.2023.3342860).
- [108] M. M. Santoni, T. Basaruddin, and K. Junus, "Convolutional neural network model based Students' engagement detection in imbalanced DAiSEE dataset," *Int. J. Adv. Comput. Sci. Appl.*, vol. 14, no. 3, pp. 617–626, 2023, doi: [10.14569/ijacsa.2023.0140371](https://doi.org/10.14569/ijacsa.2023.0140371).
- [109] S. Patil-Kashid, P. Y. M. V. R. Pawar, and K. A. S., "Facial expression recognition for low resolution images using local and global features with SVM classifier," *Int. J. Electr. Electron. Eng.*, vol. 10, no. 7, pp. 181–187, Jul. 2023, doi: [10.14445/23488379/ijeee-v10i7p116](https://doi.org/10.14445/23488379/ijeee-v10i7p116).
- [110] X. Ma, M. Xu, Y. Dong, and Z. Sun, "Automatic student engagement in online learning environment based on neural Turing machine," *Int. J. Inf. Educ. Technol.*, vol. 11, no. 3, pp. 107–111, 2021, doi: [10.18178/ijiet.2021.11.3.1497](https://doi.org/10.18178/ijiet.2021.11.3.1497).
- [111] A. Wajjira and S. Bhattacharya, "Predicting student engagement in the online learning environment," *Int. J. Web-Based Learn. Teach. Technol.*, vol. 16, no. 6, pp. 1–21, Oct. 2021, doi: [10.4018/IJWLTT.287095](https://doi.org/10.4018/IJWLTT.287095).
- [112] B. Baniere, D. Al Thani, M. Qaraqe, and B. Mansoor, "Face-based attention recognition model for children with autism spectrum disorder," *J. Healthcare Informat. Res.*, vol. 5, no. 4, pp. 420–445, Dec. 2021, doi: [10.1007/s41666-021-00101-y](https://doi.org/10.1007/s41666-021-00101-y).
- [113] A. V. Savchenko and I. A. Makarov, "Neural network model for video-based analysis of student's emotions in E-learning," *Opt. Memory Neural Netw.*, vol. 31, no. 3, pp. 237–244, Sep. 2022, doi: [10.3103/s1060992x22030055](https://doi.org/10.3103/s1060992x22030055).
- [114] T. V. George, H. S. Park, and U. Lee, "FT xtraction: Feature extraction and visualization of conversational video data for social and emotional analysis," *SoftwareX*, vol. 27, Sep. 2024, Art. no. 101827, doi: [10.1016/j.softx.2024.101827](https://doi.org/10.1016/j.softx.2024.101827).
- [115] T. S. Ashwin and R. M. R. Guddeti, "Impact of inquiry interventions on students in e-learning and classroom environments using affective computing framework," *User Model. User-Adapted Interact.*, vol. 30, no. 5, pp. 759–801, Nov. 2020, doi: [10.1007/s11257-019-09254-3](https://doi.org/10.1007/s11257-019-09254-3).
- [116] F. Ma, "Construction and evaluation of college english translation teaching model based on multimodal integration," *Appl. Math. Nonlinear Sci.*, vol. 9, no. 1, Jan. 2024, Art. no. 20241774, doi: [10.2478/amns-2024-1774](https://doi.org/10.2478/amns-2024-1774).
- [117] P. Wang, "Student sentiment analysis and classroom feedback prediction using deep learning," *Appl. Math. Nonlinear Sci.*, vol. 9, no. 1, Jan. 2024, Art. no. 20240878, doi: [10.2478/amns-2024-0878](https://doi.org/10.2478/amns-2024-0878).
- [118] J. Zaletelj and A. Košir, "Predicting students' attention in the classroom from Kinect facial and body features," *EURASIP J. Image Video Process.*, vol. 2017, no. 1, p. 80, Dec. 2017, doi: [10.1186/s13640-017-0228-8](https://doi.org/10.1186/s13640-017-0228-8).
- [119] N. Prameela, M. S. Das, and R. D. Borreo, "Multi-head network based students behaviour prediction with feedback generation for enhancing classroom engagement and teaching effectiveness," *Int. J. Inf. Technol. Comput. Sci.*, vol. 16, no. 5, pp. 81–100, Oct. 2024, doi: [10.5815/ijitcs.2024.05.06](https://doi.org/10.5815/ijitcs.2024.05.06).
- [120] A. Rajae, R. Amina, and I. E. H. El Hassane, "An improved student's facial emotions recognition method using transfer learning," *Indonesian J. Electr. Eng. Comput. Sci.*, vol. 36, no. 2, p. 1199, Nov. 2024, doi: [10.11591/ijeecs.v36.i2.pp1199-1208](https://doi.org/10.11591/ijeecs.v36.i2.pp1199-1208).
- [121] D. Evangeline and A. Parkavi, "Facial emotion recognition of online learners using a hybrid deep learning model," *Int. J. Intell. Eng. Syst.*, vol. 17, no. 6, pp. 735–751, Dec. 2024, doi: [10.22266/ijies2024.1231.56](https://doi.org/10.22266/ijies2024.1231.56).

- [122] S. Wang, A. S. Shibghatullah, K. H. Keoy, and J. Iqbal, "YOLOv5 based student engagement and emotional states detection in E-classes," *J. Robot., Netw. Artif. Life*, vol. 10, no. 4, pp. 357–361, Mar. 2024.
- [123] S. Hasegawa, A. Hirako, X. Zheng, S. N. Karimah, K. Ota, and T. Unoki, "Learner's mental state estimation with PC built-in camera," in *Learning and Collaboration Technologies. Human and Technology Ecosystems* (Lecture Notes in Computer Science), vol. 12206. Cham, Switzerland: Springer, 2020, pp. 165–175, doi: [10.1007/978-3-030-50506-6_12](https://doi.org/10.1007/978-3-030-50506-6_12).
- [124] Y. Chen, J. Zhou, Q. Gao, J. Gao, and W. Zhang, "MDNN: Predicting student engagement via gaze direction and facial expression in collaborative learning," *Comput. Model. Eng. Sci.*, vol. 136, no. 1, pp. 381–401, 2023, doi: [10.32604/cmes.2023.023234](https://doi.org/10.32604/cmes.2023.023234).
- [125] Z. Abdelhadi, M. Naseif, W. Alhejali and A. Elhayek, "TeacherEye: An AI-powered system for monitoring student engagement in online education," in *Proc. 22nd Int. Learn. Technol. Conf. (L&T)*, Jeddah, Saudi Arabia, 2025, pp. 25–30, doi: [10.1109/LT64002.2025.10940905](https://doi.org/10.1109/LT64002.2025.10940905).
- [126] P. E. Griffiths, "III. Basic emotions, complex emotions, machiavellian emotions," *Roy. Inst. Philosophy Suppl.*, vol. 52, pp. 39–67, Mar. 2003, doi: [10.1017/s1358246100007888](https://doi.org/10.1017/s1358246100007888).
- [127] P. Ekman, W. V. Friesen, M. O'Sullivan, A. Chan, I. Diacoyanni-Tarlatzis, K. Heider, R. Krause, W. A. LeCompte, T. Pitcairn, P. E. Ricci-Bitti, K. Scherer, M. Tomita, and A. Tzavaras, "Universals and cultural differences in the judgments of facial expressions of emotion," *J. Personality Social Psychol.*, vol. 53, no. 4, pp. 712–717, 1987, doi: [10.1037/0022-3514.53.4.712](https://doi.org/10.1037/0022-3514.53.4.712).
- [128] J. S. Bazargani, A. Sadeghi-Niaraki, and S.-M. Choi, "Design, implementation, and evaluation of an immersive virtual reality-based educational game for learning topology relations at schools: A case study," *Sustainability*, vol. 13, no. 23, p. 13066, Nov. 2021, doi: [10.3390/su132313066](https://doi.org/10.3390/su132313066).
- [129] J. S. Bazargani, A. Sadeghi-Niaraki, and S.-M. Choi, "A survey and framework for education in the metaverse," *IEEE Access*, vol. 13, pp. 33231–33245, 2025, doi: [10.1109/ACCESS.2025.3543496](https://doi.org/10.1109/ACCESS.2025.3543496).
- [130] X. Huang, D. Zou, G. Cheng, and H. Xie, "A systematic review of AR and VR enhanced language learning," *Sustainability*, vol. 13, no. 9, p. 4639, Apr. 2021, doi: [10.3390/su13094639](https://doi.org/10.3390/su13094639).



IKRAM QARBAL received the bachelor's degree from the Faculty of Sciences and Techniques, Mohammedia, Morocco, in 2021, and the master's degree from the Faculty of Sciences, Casablanca, Morocco, in 2023. She is currently pursuing the Ph.D. degree in data science with the Laboratory of Information Technology and Modeling, Faculty of Sciences Ben M'Sick, Hassan II University of Casablanca, Morocco. Her research interests include machine learning, deep learning, AI, and computer vision in the field of e-learning.



NAWAL SAEL received the Engineering degree in software engineering from ENSIAS, in 2002. She has been a Teacher-Researcher, since 2012, an Authorized Professor, since 2014, and a Professor of higher education with the Department of Mathematics and Computer Science, Faculty of Sciences Ben M'Sick, Casablanca, Morocco, since 2020. Her research interests include data mining, machine learning, deep learning, and the Internet of Things.



SARA OUAHABI is currently a Habilitated Professor (PH) with the Mathematics and Computer Science Department, Hassan II University of Casablanca, and a member with the Computer Science and Information Processing Laboratory, Faculty of Science Ben M'Sick. Her research interests include artificial intelligence, computer security, the semantic web, e-learning, computer communications (networks), and the Internet of Things (IoT). She conducts cutting-edge research in these areas while actively engaging in higher education to educate the next generation of computer professionals.

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